

A MINI PROJECT REPORT ON
BIRD VS DRONE IMAGE CLASSIFICATION

Submitted in Partial Fulfillment of the Requirements for The Award of Degree of

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CERTIFICATE

This is to certify that the project report entitled " BIRD vs DRONE IMAGE CLASSIFICATION ",being submitted by K.Ashwitha (20C31A0527), G.Aravind Kumar (20C31A0520), E. Akhila (20C31A0515), J. Shivudu (20C31A0523) of Bachelor of Technology IV-I Sem has satisfactorily completed the Mini Project entitled “**BIRD VS DRONE IMAGE CLASSIFICATION**”, as part of curriculum in computer Science of Engineering during academic year 2023-24.

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ABSTRACT

Adopting effective techniques to automatically detect and identify small drones is a very compelling need for a number of different stakeholders in both the public and private sectors. This work presents three different original approaches that competed in a grand challenge on the "Drone vs. Bird" detection problem.

The goal is to detect one or more drones appearing at some point in video sequences where birds and other distractor objects may be also present, together with motion in background or foreground. Algorithms should raise an alarm and provide a position estimate only when a drone is present, while not issuing alarms on birds, nor being confused by the rest of the scene.

In particular, three original approaches based on different deep learning strategies are proposed and compared on a real-world dataset provided by a consortium of universities and research centers, under the 2020 edition of the Drone vs. Bird Detection Challenge. Results show that there is a range of difficulty among different test sequences, depending on the size and the shape visibility of the drone in the sequence, while sequences recorded by a moving camera and very distant drones are the most challenging ones. The performance comparison reveals that the different approaches perform somewhat complementary, in terms of correct detection rate, false alarm rate, and average precision.

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1. MOTIVATION

Drone detection has become an essential task in object detection as drone costs have decreased and drone technology has improved. It is, however, difficult to detect distant drones when there is weak contrast, long range, and low visibility.

Initially used for military purposes, drones are now used in a variety of other fields as well, such as traffic and weather monitoring, and smart agriculture monitoring, just to name a few. In addition, since the COVID-19 pandemic, drones have been increasingly used to deliver groceries and medical supplies autonomously and enforce social segregation. Nowadays, small quadcopters can be easily purchased on the Internet at low prices, which brings unprecedented opportunities but poses several threats regarding safety, privacy, and security.

This project aimed to address the technical issues of discriminating between drones and birds. In practice, drones can be easily confused with birds, particularly at long distances, which makes the surveillance task even more challenging. The use of video analytics can solve the issue, but effective algorithms are needed, which can operate under unfavorable conditions, such as weak contrast, long-range, low visibility, etc.

2. PROJECT OBJECTIVES

The aim of the project is to classify and predict what a small flying object in the sky is by capturing its image and passing to an Artificial Neural Network (ANN) whose functionality is to figure out and predict if that unidentified flying object that is captured is bird (no potential threat) or a drone (a potential threat) by simply passing the image to the Neural Network.

This application can be used in areas where flying drones are restricted due to security and protection purposes and if the application identifies the flying object correctly as a drone, then necessary action can be taken to take down the drone, preserve the security and eliminate any potential threat.

Innocent flying birds that are not of any threat to any individual or a group of people will not be harmed due to this effort and they will be flying as usual.

3. PROPOSED WORK

3.1. Functional Requirements

- **Python:**
 - Python is the basis of the program that we wrote. It utilizes many of the Python libraries.
- **Libraries:**
 - **TensorFlow:** It is a software library or framework, used to implement machine learning and deep learning concepts in the easiest manner. It includes programming support of deep neural networks and machine learning techniques. It includes a highly scalable feature of computation with various data sets.
 - **Keras:** It runs on top of open-source machine libraries like TensorFlow. It is based on a minimal structure that provides a clean and easy way to create deep learning models based on TensorFlow or Theano. It is designed to quickly define deep learning models.
- **Operating System**

3.2. Non-Functional Requirements

- **Usability:** It should be easy for the user, and easy to learn, operate through interaction with a system,
- **Scalability:** Application performance should be consistent while processing different kinds of images.
- **Flexibility:** Any hardware and software configuration application should work fine.
- **Efficiency:** Tool should be able to predict the tumor efficiently.

3.3. Workflow

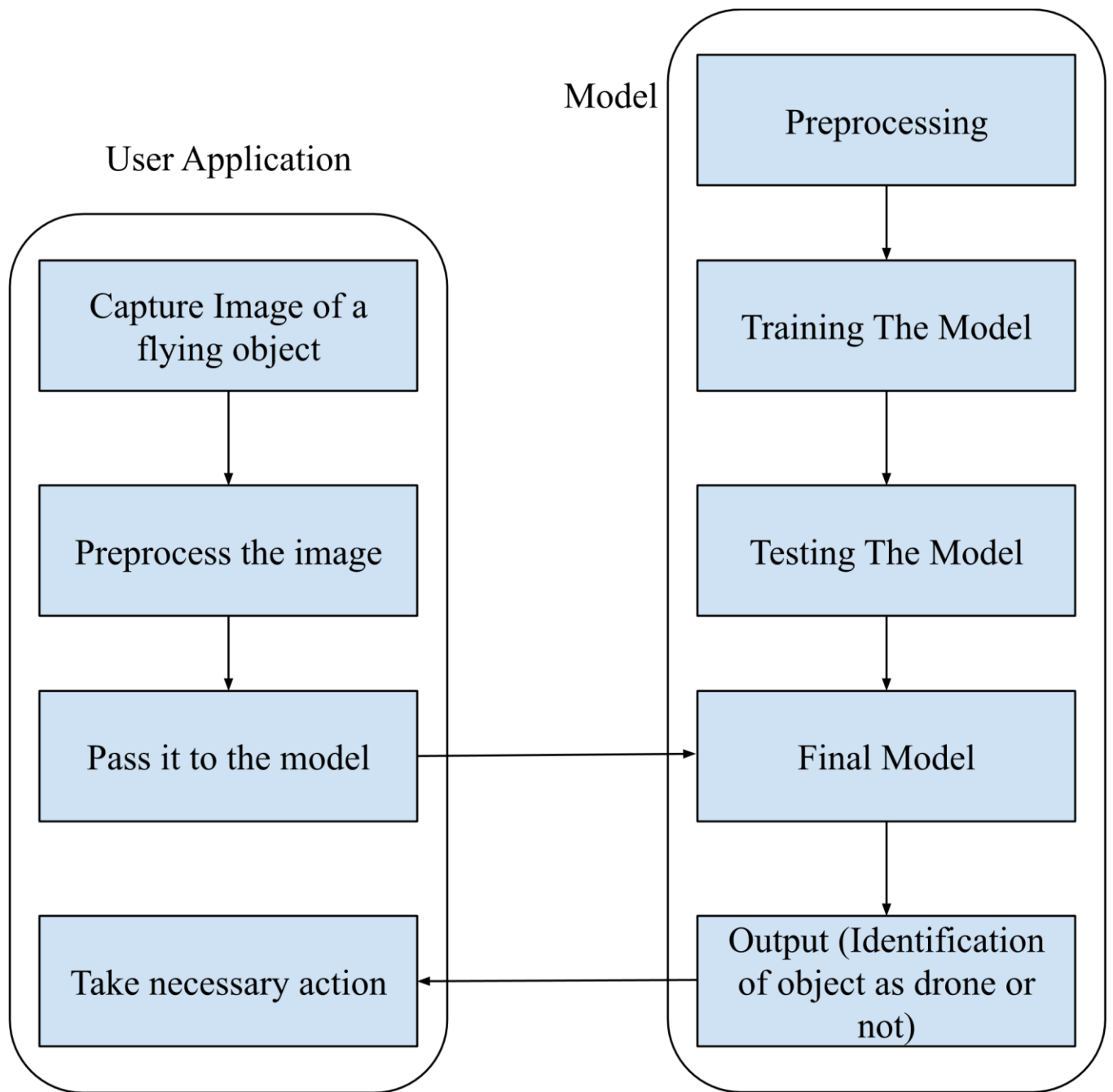


Fig 3.1 Workflow of the Project

4. INTRODUCTION

4.1. Deep Learning

Deep learning is a subset of machine learning, which is essentially a neural network with three or more layers. These neural networks attempt to simulate the behavior of the human brain—albeit far from matching its ability—allowing it to “learn” from large amounts of data. While a neural network with a single layer can still make approximate predictions, additional hidden layers can help to optimize and refine for accuracy.

Deep learning neural networks, or artificial neural networks, attempts to mimic the human brain through a combination of data inputs, weights, and bias. These elements work together to accurately recognize, classify, and describe objects within the data.

Deep learning drives many artificial intelligence (AI) applications and services that improve automation, performing analytical and physical tasks without human intervention. Deep learning technology lies behind everyday products and services (such as digital assistants, voice-enabled TV remotes, and credit card fraud detection) as well as emerging technologies (such as self-driving cars).

4.2. TensorFlow

TensorFlow is a software library or framework, designed by the Google team to implement machine learning and deep learning concepts in the easiest manner. It combines the computational algebra of optimization techniques for easy calculation of many mathematical expressions.

TensorFlow is well-documented and includes plenty of machine learning libraries. It offers a few important functionalities and methods for the same. TensorFlow is

a "Google" product. It includes a variety of machine learning and deep learning algorithms.

TensorFlow can train and run deep neural networks for handwritten digit classification, image recognition, word embedding and creation of various sequence models.

4.3. Feed Forward Neural Network

A Feed Forward Neural Network is an artificial neural network in which the connections between nodes do not form a cycle. The opposite of a feed forward neural network is a recurrent neural network, in which certain pathways are cycled. The feed forward model is the simplest form of neural network as information is only processed in one direction. While the data may pass through multiple hidden nodes, it always moves in one direction and never backwards.

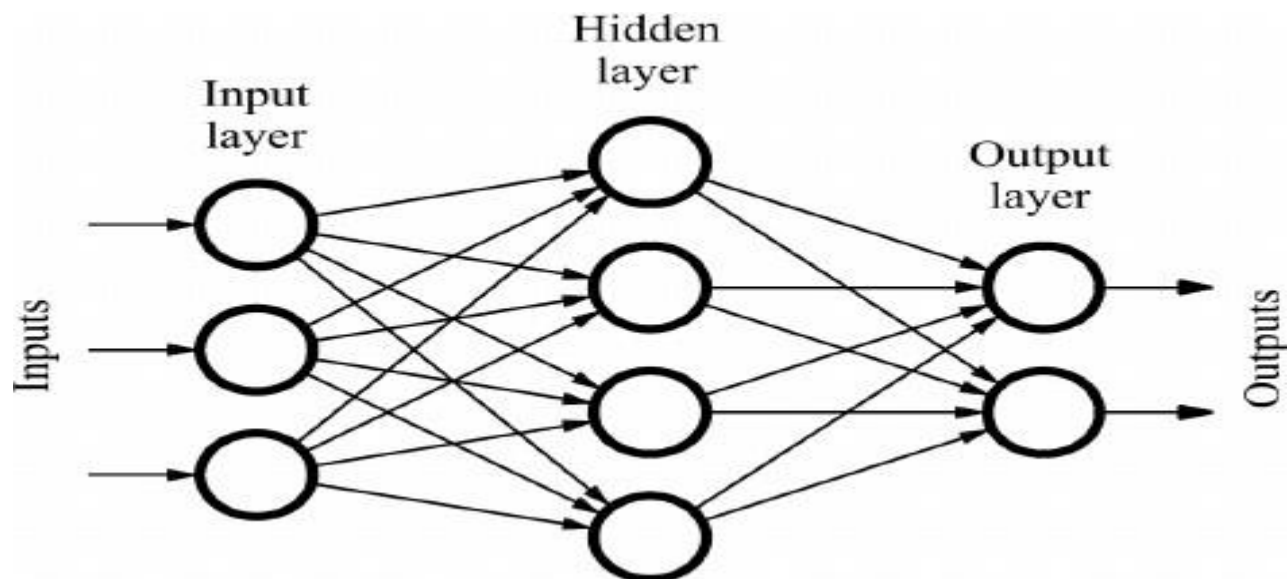


Fig 4.1 Basic Structure of Neural Network

4.3.1 How does a Feed Forward Neural Network work?

A Feed Forward Neural Network is commonly seen in its simplest form as a single layer perceptron. In this model, a series of inputs enter the layer and are multiplied by the weights. Each value is then added together to get a sum of the weighted input values. If the sum of the values is above a specific threshold, usually set at zero, the value produced is often 1, whereas if the sum falls below the threshold, the output value is -1. The single layer perceptron is an important model of feed forward neural networks and is often used in classification tasks. Furthermore, single layer perceptrons can incorporate aspects of machine learning. Using a property known as the delta rule, the neural network can compare the outputs of its nodes with the intended values, thus allowing the network to adjust its weights through training in order to produce more accurate output values. This process of training and learning produces a form of a gradient descent. In multi-layered perceptrons, the process of updating weights is nearly analogous, however the process is defined more specifically as back-propagation. In such cases, each hidden layer within the network is adjusted according to the output values produced by the final layer.

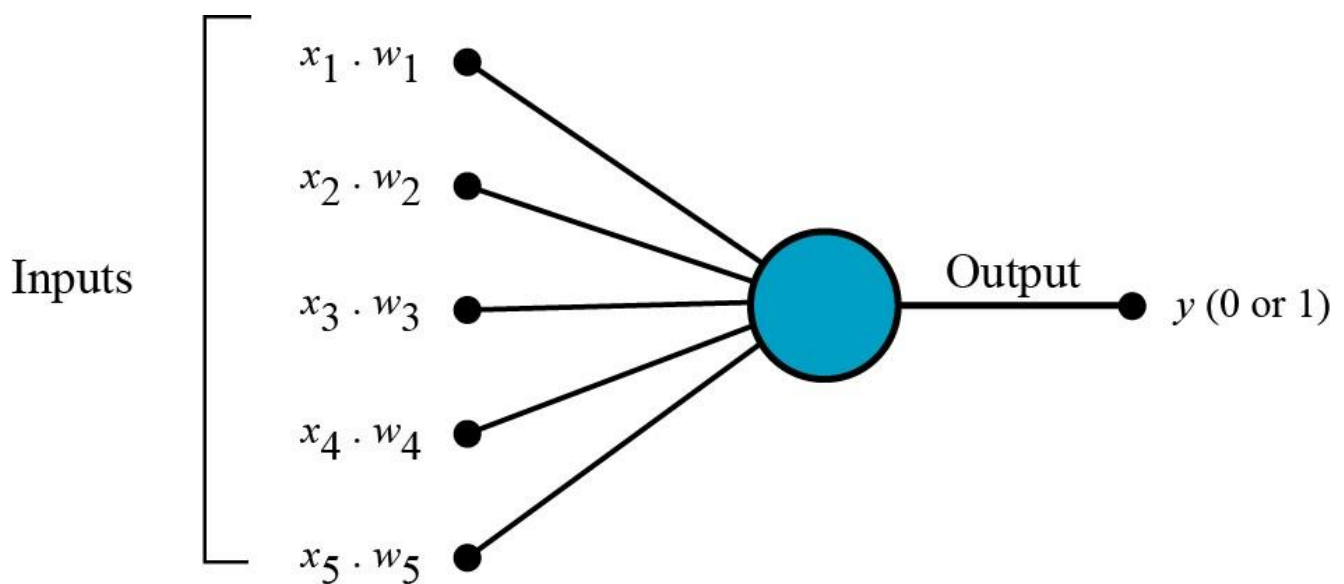


Fig 4.2 Anatomy of a Neuron

4.3.2 Applications of Feed Forward Neural Networks

While Feed Forward Neural Networks are fairly straightforward, their simplified architecture can be used as an advantage in particular machine learning applications. For example, one may set up a series of feed forward neural networks with the intention of running them independently from each other, but with a mild intermediary for moderation. Like the human brain, this process relies on many individual neurons in order to handle and process larger tasks. As the individual networks perform their tasks independently, the results can be combined at the end to produce a synthesized, and cohesive output.

4.4 Convolutional Neural Network

A Convolutional Neural Network (ConvNet/CNN) is a Deep Learning algorithm which can take in an input image, assign importance (learnable weights and biases) to various aspects/objects in the image and be able to differentiate one from the other. The pre-processing required in a ConvNet is much lower as compared to other classification algorithms. While in primitive methods filters are hand-engineered, with enough training, ConvNets have the ability to learn these filters/characteristics.

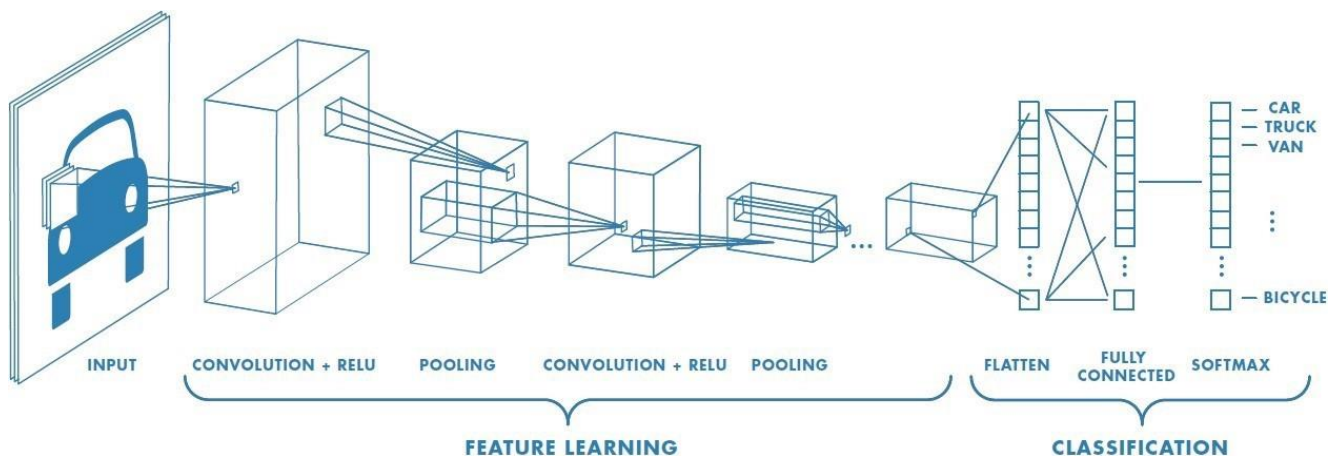


Fig 4.3 Working of a Convolutional Neural Network

The architecture of a ConvNet is analogous to that of the connectivity pattern of Neurons in the Human Brain and was inspired by the organization of the Visual Cortex. Individual neurons respond to stimuli only in a restricted region of the visual field known as the Receptive Field. A collection of such fields overlap to cover the entire visual area.

4.4.1 Feature Learning, Layers, and Classification

Like other neural networks, a CNN is composed of an input layer, an output layer, and many hidden layers in between.

These layers perform operations that alter the data with the intent of learning features specific to the data. Three of the most common layers are:

- **Convolution** puts the input images through a set of convolutional filters, each of which activates certain features from the images.
- **Rectified linear unit (ReLU)** allows for faster and more effective training by mapping negative values to zero and maintaining positive values. This is sometimes referred to as activation, because only the activated features are carried forward into the next layer.
- **Pooling** simplifies the output by performing nonlinear downsampling, reducing the number of parameters that the network needs to learn.

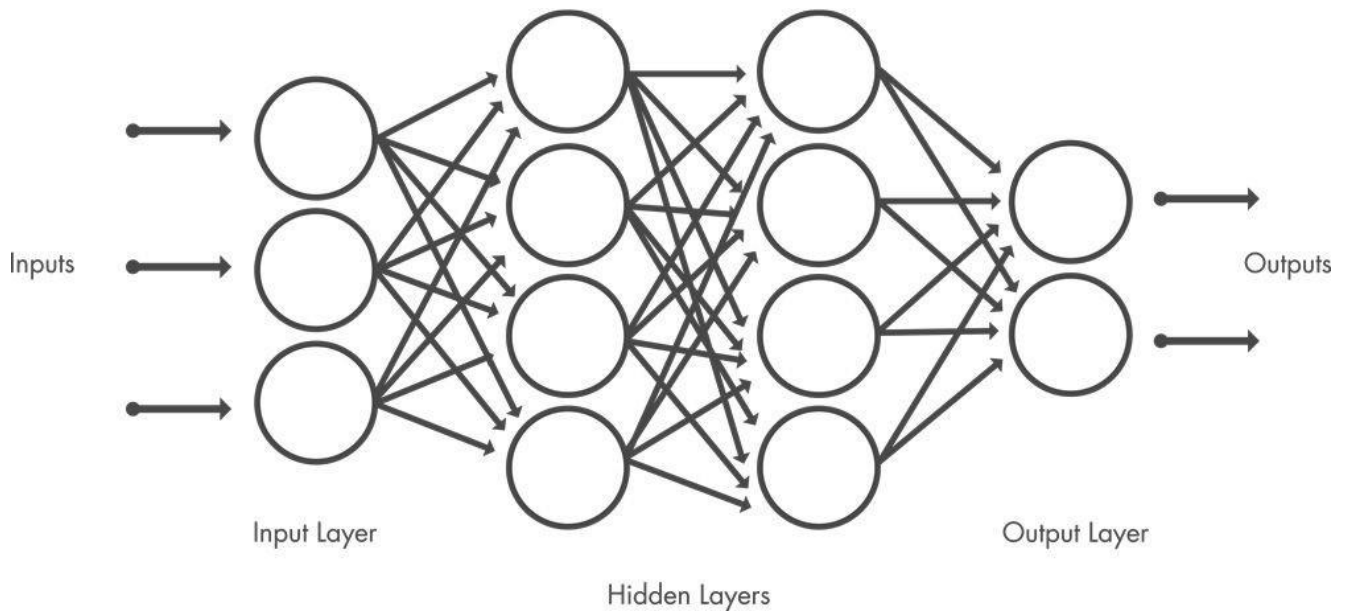


Fig 4.4 A Simple Dense Neural Network

4.4.2 Uses of CNNs

Using CNNs for deep learning is popular due to three important factors:

- CNNs eliminate the need for manual feature extraction—the features are learned directly by the CNN.
- CNNs produce highly accurate recognition results.
- CNNs can be retrained for new recognition tasks, enabling you to build on pre-existing networks.

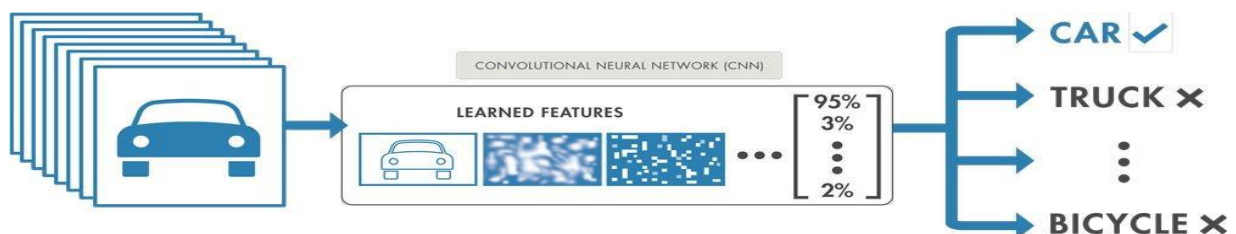


Fig 4.5 Classification using CNN

4.4.3 Applications of CNN

CNNs provide an optimal architecture for uncovering and learning key features in image and time-series data. CNNs are a key technology in applications such as:

- **Medical Imaging:** CNNs can examine thousands of pathology reports to visually detect the presence or absence of cancer cells in images.
- **Audio Processing:** Keyword detection can be used in any device with a microphone to detect when a certain word or phrase is spoken - ('Hey Siri!'). CNNs can accurately learn and detect the keyword while ignoring all other phrases regardless of the environment.
- **Stop Sign Detection:** Automated driving relies on CNNs to accurately detect the presence of a sign or other object and make decisions based on the output.
- **Synthetic Data Generation:** Using Generative Adversarial Networks (GANs), new images can be produced for use in deep learning applications including face recognition and automated driving.

4.5 Recurrent Neural Network

A Deep Learning approach for modeling sequential data is Recurrent Neural Networks (RNN). RNNs were the standard suggestion for working with sequential data before the advent of attention models. Specific parameters for each element of the sequence may be required by a deep feedforward model. It may also be unable to generalize to variable-length sequences.

Recurrent Neural Networks use the same weights for each element of the sequence, decreasing the number of parameters and allowing the model to

generalize to sequences of varying lengths. RNNs generalize to structured data other than sequential data, such as geographical or graphical data, because of its design.

RNNs are a type of neural network that can be used to model sequence data. RNNs, which are formed from feedforward networks, are similar to human brains in their behavior. Simply said, recurrent neural networks can anticipate sequential data in a way that other algorithms can't.

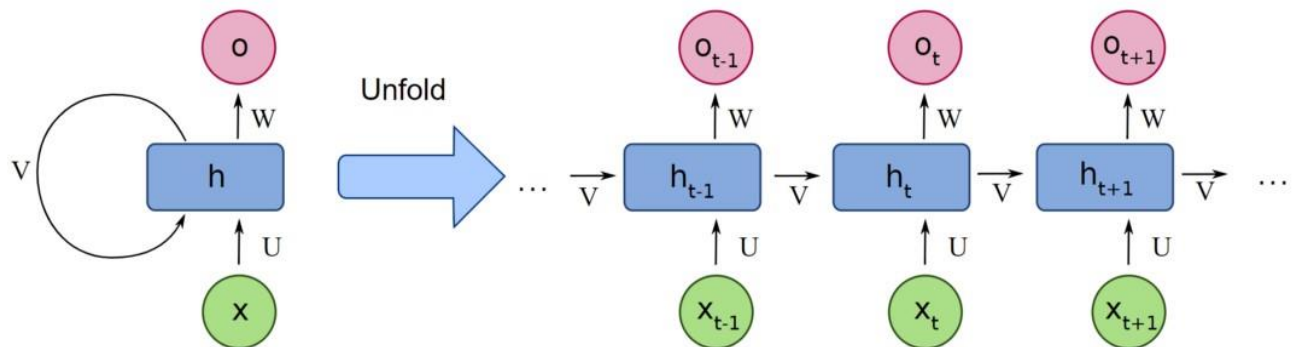


Fig 4.6 Sequence of RNN inputs

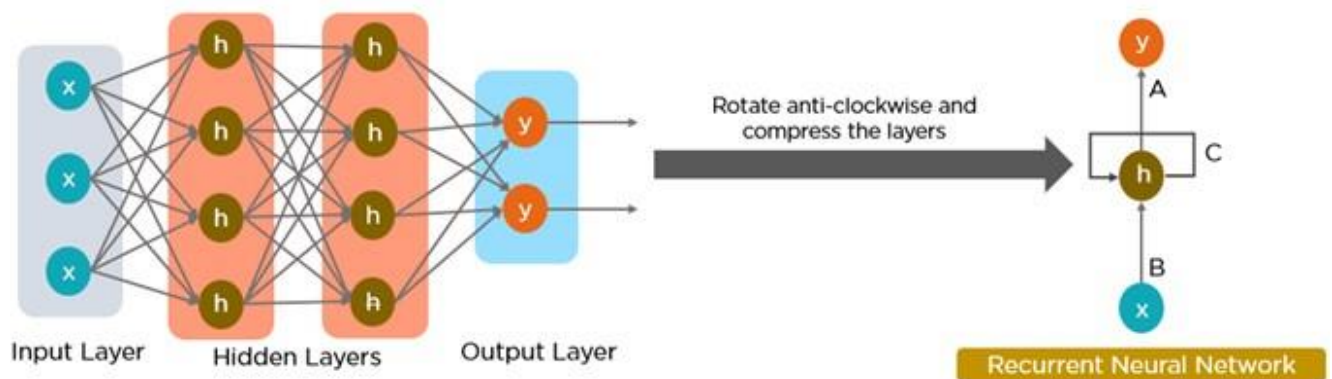


Fig 4.7 Comparison of Dense Network and Recurrent Network

All of the inputs and outputs in standard neural networks are independent of one another, however in some circumstances, such as when predicting the next word of a phrase, the prior words are necessary, and so the previous words must be remembered.

As a result, RNN was created, which used a Hidden Layer to overcome the problem. The most important component of RNN is the Hidden state, which remembers specific information about a sequence.

RNNs have a Memory that stores all information about the calculations. It employs the same settings for each input since it produces the same outcome by performing the same task on all inputs or hidden layers.

4.6 Image Classification

Image classification refers to the task of extracting information classes from a multiband raster image. The resulting raster from image classification can be used to create thematic maps. Depending on the interaction between the analyst and the computer during classification, there are two types of classification: supervised and unsupervised.

With the ArcGIS Spatial Analyst extension, there is a full suite of tools in the Multivariate toolset to perform supervised and unsupervised classification (see An overview of the Multivariate toolset). The classification process is a multi-step workflow, therefore, the Image Classification toolbar has been developed to provide an integrated environment to perform classifications with the tools. Not only does the toolbar help with the workflow for performing unsupervised and supervised classification, it also contains additional functionality for analyzing input data, creating training samples and signature files, and determining the quality of the training samples and signature files. The recommended way to perform classification and multivariate analysis is through the Image Classification toolbar.

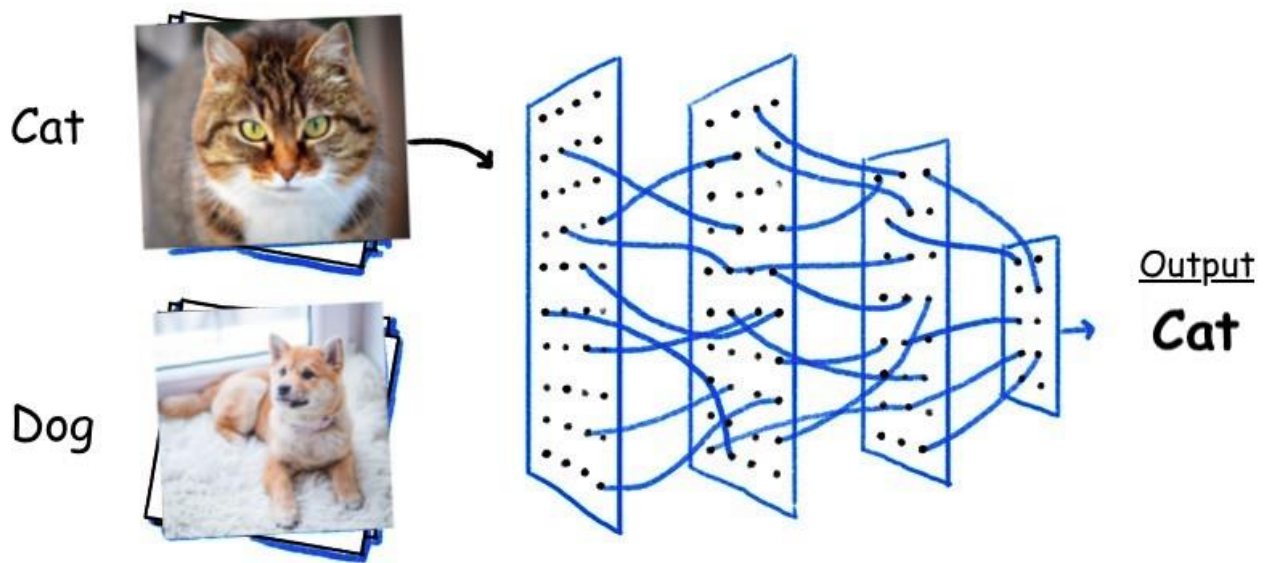


Fig 4.8 Example of Binary Image Classification

4.6.1 Supervised classification

Supervised classification uses the spectral signatures obtained from training samples to classify an image. With the assistance of the Image Classification toolbar, you can easily create training samples to represent the classes you want to extract. You can also easily create a signature file from the training samples, which is then used by the multivariate classification tools to classify the image.

4.6.2 Unsupervised classification

Unsupervised classification finds spectral classes (or clusters) in a multiband image without the analyst's intervention. The Image Classification toolbar aids in unsupervised classification by providing access to the tools to create

the clusters, capability to analyze the quality of the clusters, and access to classification tools.

4.7. Binary Image Classification

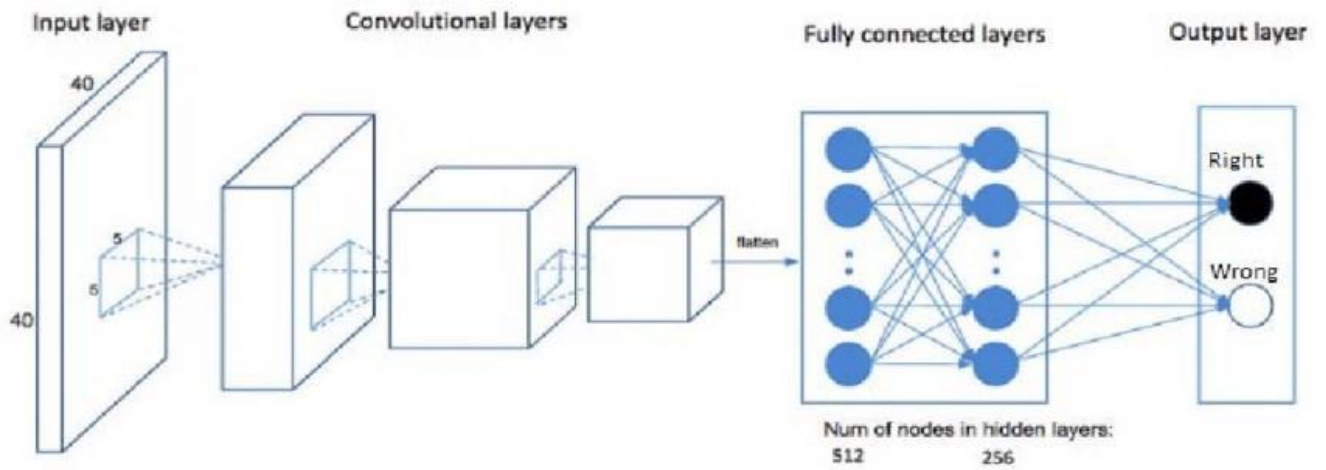


Fig 4.9 Binary Image Classification using CNN

5. CNN ARCHITECTURE

5.1. Convolutional Neural Network

For these reasons, the Convolutional Neural Network takes a different approach, mimicking the way we perceive our environment with our eyes. When we see an image, we automatically divide it into many small sub-images and analyze them one by one. By assembling these sub-images, we process and interpret the image. How can this principle be implemented in a Convolutional Neural Network?

The work happens in the so-called convolution layer. To do this, we define a filter that determines how large the partial images we are looking at should be, and a step length that decides how many pixels we continue between calculations, i.e. how close the partial images are to each other. By taking this step, we have greatly reduced the dimensionality of the image.

The next step is the pooling layer. From a purely computational point of view, the same thing happens here as in the convolution layer, with the difference that we only take either the average or maximum value from the result, depending on the application. This preserves small features in a few pixels that are crucial for the task solution.

Finally, there is a fully-connected layer, as we already know it from the normal neural networks. Now that we have greatly reduced the dimensions of the image, we can use the tightly meshed layers. Here, the individual sub-images are linked again in order to recognize the connections and to carry out the classification.

5.1.1. Convolutional Layer

In the first step, we want to reduce the dimensions of the 4 x 4 x 3 image. For this purpose, we define a filter with the dimension 2 x 2 for each color. In addition, we want a step length of 1, i.e. after each calculation step, the filter should be moved forward by exactly one pixel. This will not reduce the dimension as much, but details of the image will be preserved. If we migrate a 4x4 matrix with a 2 x 2 and advance one column or one row in each step, our Convolutional Layer will have a 3 x 3 matrix as output. The individual values of the matrix are calculated by taking the scalar product of the 2 x 2 matrices, as shown in the graphic.

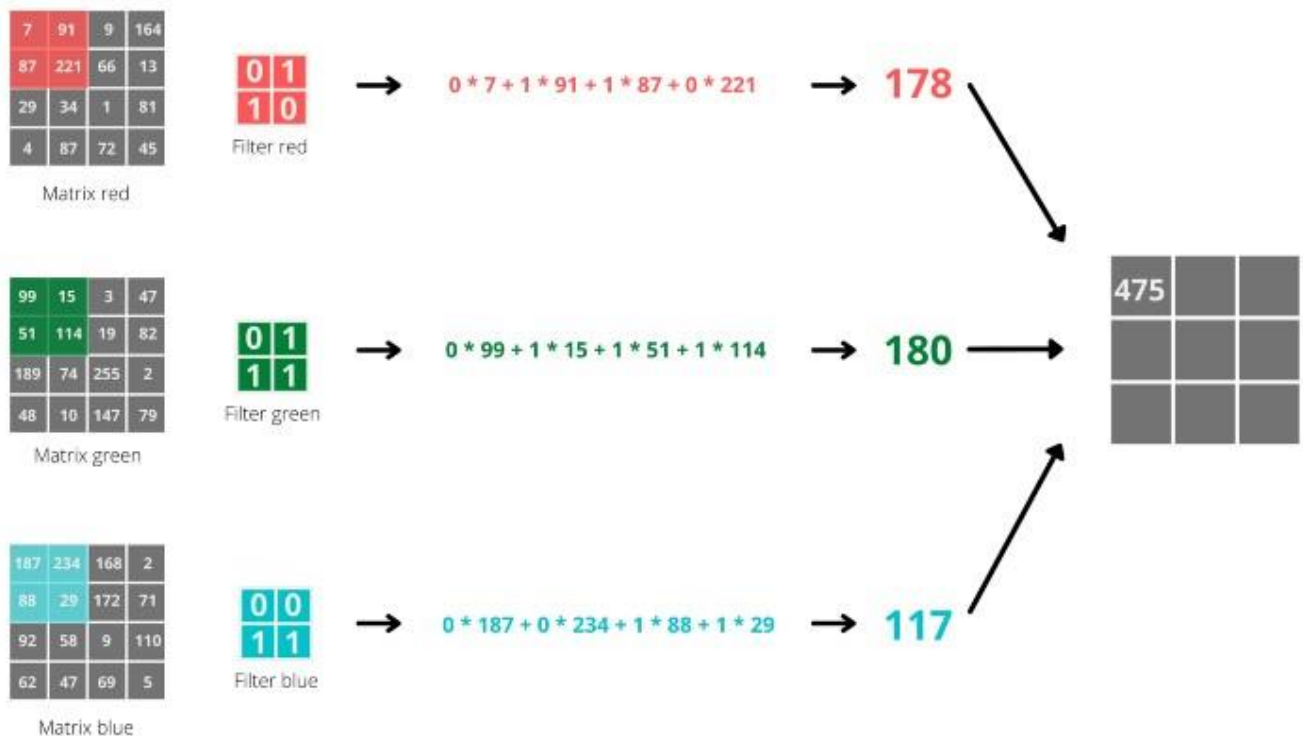


Fig 5.1 Convolutional Operations

5.1.2. Pooling Layer

The (Max) Pooling Layer takes the 3 x 3 matrix of the convolution layer as input and tries to reduce the dimensionality further and additionally take the important features in the image. We want to generate a 2 x 2 matrix as the output of this layer, so we divide the input into all possible 2 x 2 partial matrices and search for the highest value in these fields. This will be the value in the field of the output matrix. If we were to use the average pooling layer instead of a max-pooling layer, we would calculate the average of the four fields instead.

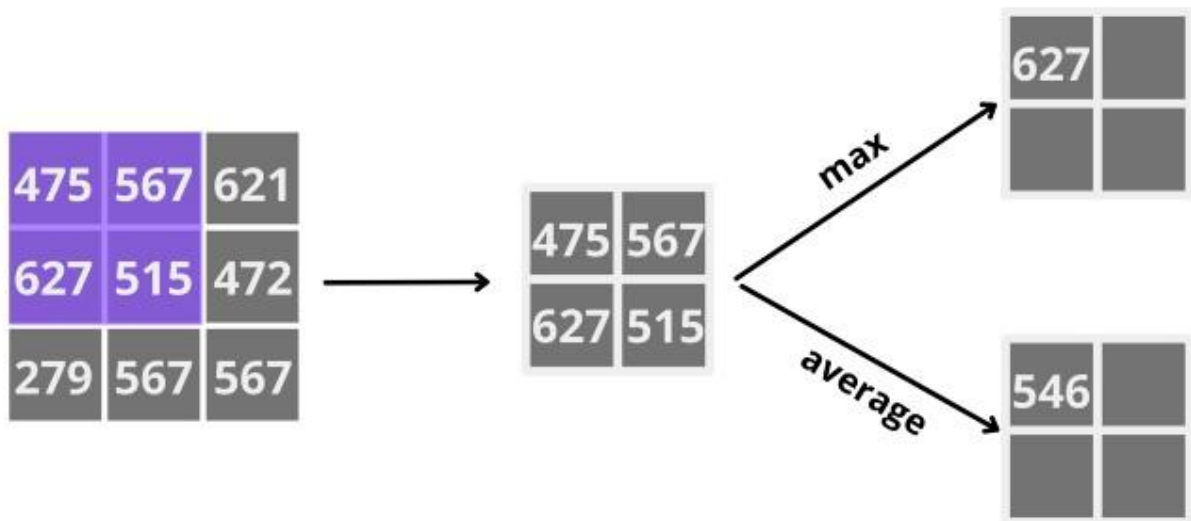


Fig 5.2 Max Pooling Operations

The pooling layer also filters out noise from the image, i.e. elements of the image that do not contribute to the classification. For example, whether the dog is standing in front of a house or in front of a forest is not important at first.

5.1.3 Fully-Connected Layer

The fully-connected layer now does exactly what we intended to do with the whole image at the beginning. We create a neuron for each entry in the smaller 2×2 matrix and connect them to all neurons in the next layer. This gives us significantly fewer dimensions and requires fewer resources in training. This layer then finally learns which parts of the image are needed to make the classification dog or non-dog. If we have images that are much larger than our $5 \times 5 \times 3$ example, it is of course also possible to set the convolution layer and pooling layer several times in a row before going into the fully-connected layer. This way you can reduce the dimensionality far enough to reduce the training effort.

6. DATASET

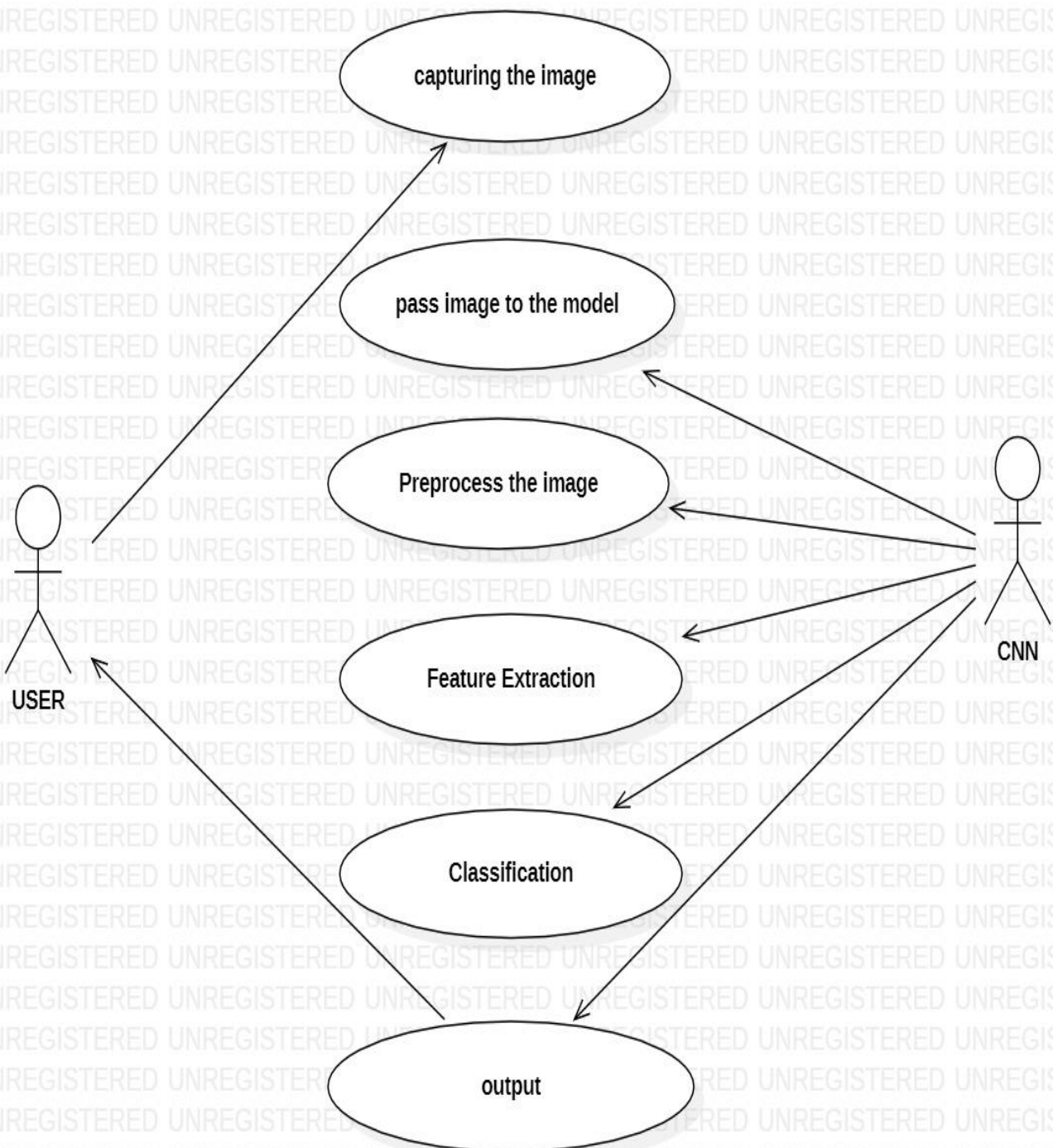
TensorFlow has a wide variety of datasets that we can download and use with just a few lines of code. This is especially helpful when you want to test new models and their implementation and therefore do not want to search for appropriate data for a long time. In addition, Google also offers a dataset search, with which one can find a suitable dataset within a few clicks.

For our exemplar Convolutional Neural Network, we use the Bird vs. Drone Image dataset, which is available in Kaggle. The dataset contains a total of 828 images in color, divided into two different image classes, i.e. birds and drones. We then split 80% of this dataset into a training dataset (660) and 20% into a test dataset (166). We note that this is a perfect training dataset as each class contains exactly 6,000 images. In classification models, we must always make sure that every class is included in the dataset an equal number of times, if possible.

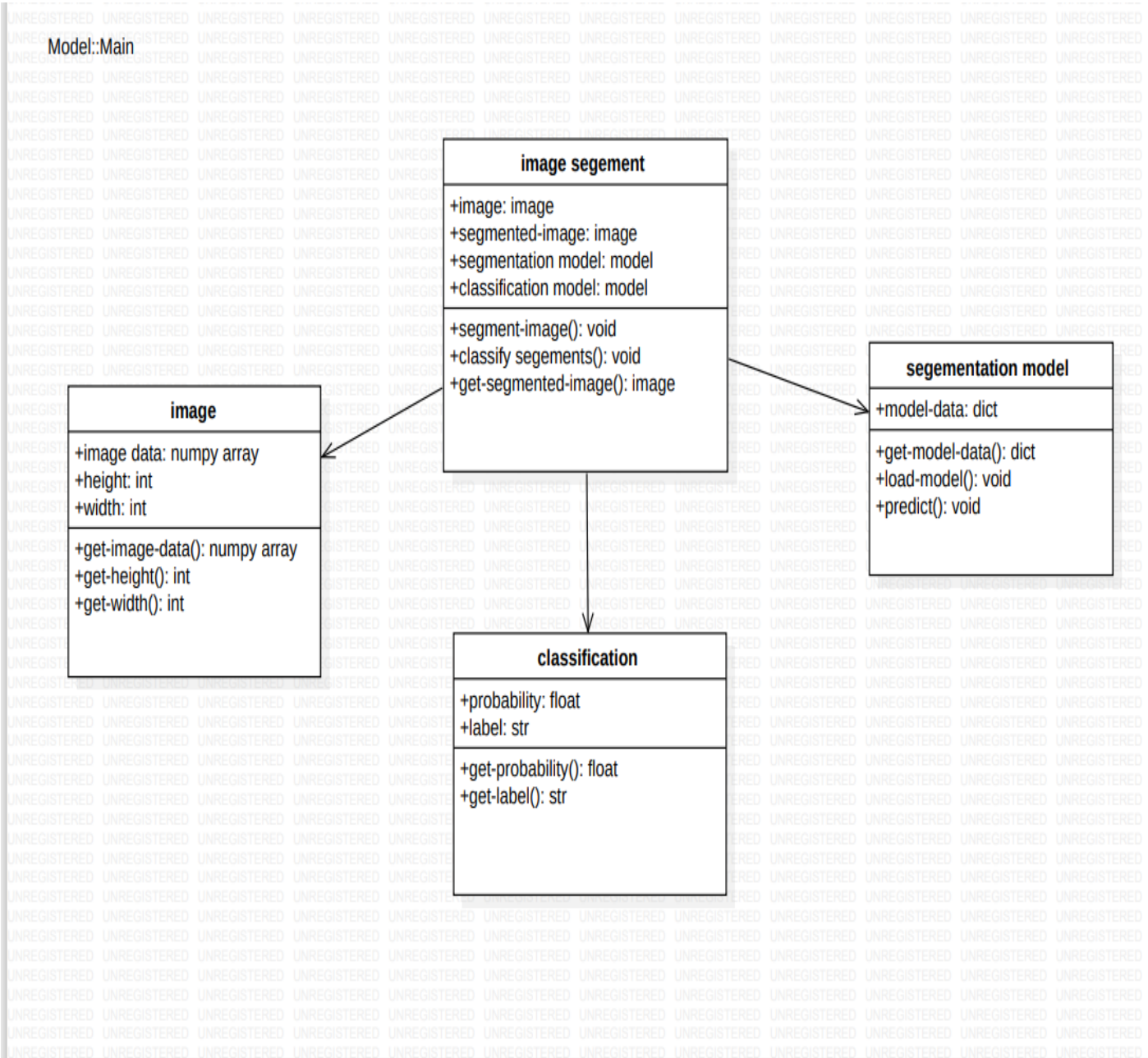
Each of these images is random in size and has a random aspect ratio. In order to make the images compatible to be fed into the neural network, which expects all the images to be of the same shape, we scale the images to all be 224 x 224 pixels. The pixels in turn have a value between 0 and 255, where each number represents a color code. Therefore, we divide each pixel value by 255 so that we normalize the pixel values to the range between 0 and 1.

7.UML DIAGRAMS

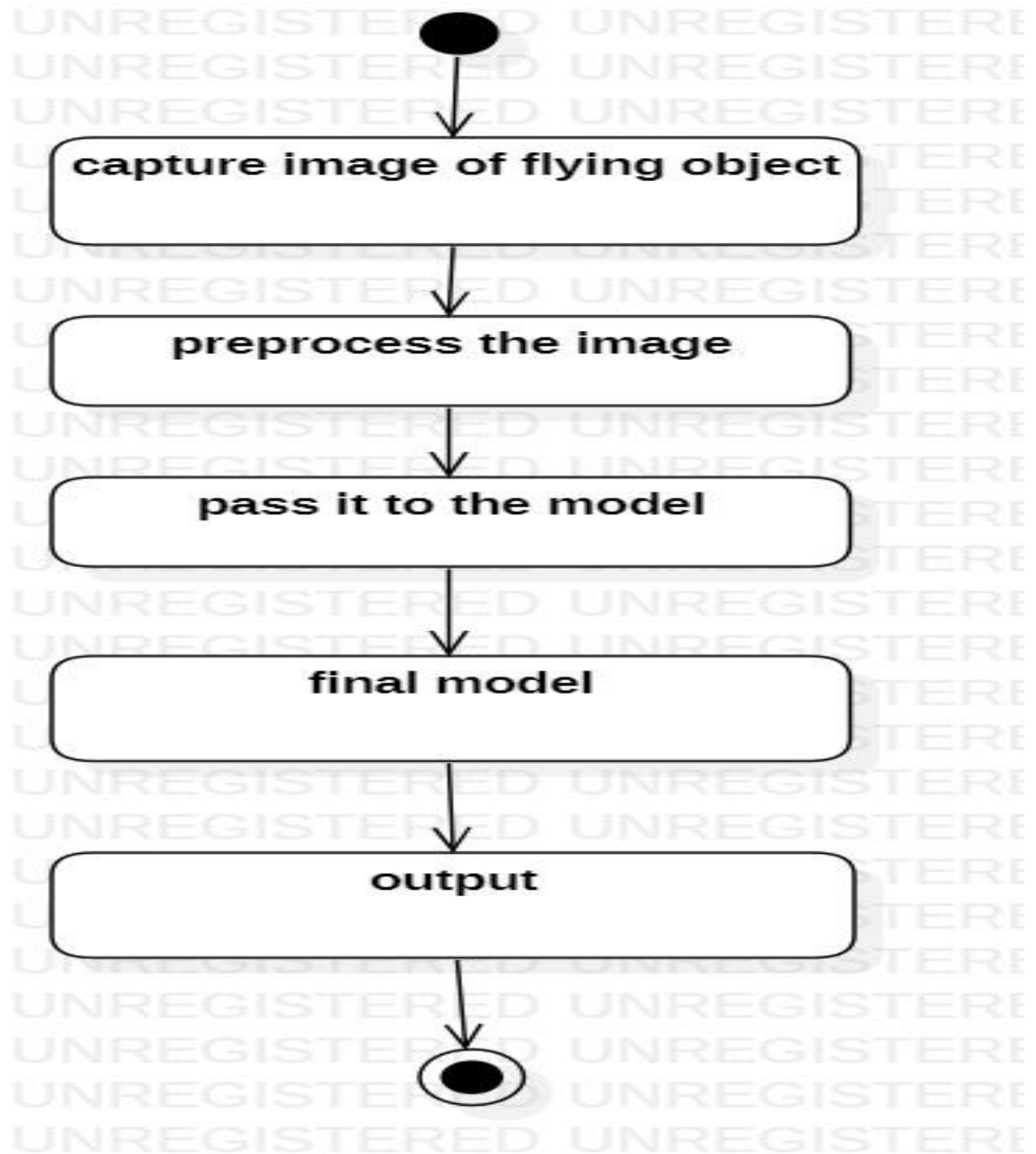
7.1 Use Case Diagram



7.2 Class Diagram



7.3 Activity Diagram



8. BUILD A CONVOLUTIONAL NEURAL NETWORK

In Tensorflow we can now build the Convolutional Neural Network by defining the sequence of each layer. Since we are dealing with relatively small images we will use the stack of Convolutional Layer and Max Pooling Layer thrice. The images have, as we already know, 224 height dimensions, 224 width dimensions, and 3 color channels (red, green, blue).

The first Convolutional Layer uses 16 filters, the second uses 32 filters and the final Convolutional Layer uses 64 filters with each layer having a 3×3 kernel as a filter and the Max Pooling Layer searches for the maximum value within a 2×2 matrix. If you are still unfamiliar with this approach, you are welcome to read the theoretical article on Convolutional Neural Networks again.

Now we have sufficiently reduced the dimensions of the images and can add one more hidden layer with a total of 512 neurons before the model ends in the output layer with the one neuron.

The model with a total of 8,243,345 parameters is now ready to be built and trained.

9. TENSORFLOW LAYERS

9.1. Conv2D

2D convolution layer (e.g. spatial convolution over images)

The convolutional neuron performs an elementwise dot product with a unique kernel and the output of the previous layer's corresponding neuron. This will yield as many intermediate results as there are unique kernels. The convolutional neuron is the result of all of the intermediate results summed together with the learned bias.

```
tf.keras.layers.Conv2D( filters,
    kernel_size,  strides=(1, 1),
    padding='valid',  data_format=None,
    dilation_rate=(1, 1),  groups=1,
    activation=None,  use_bias=True,
    kernel_initializer='glorot_uniform',
    bias_initializer='zeros',
    kernel_regularizer=None,
    bias_regularizer=None,
    activity_regularizer=None,
    kernel_constraint=None,
    bias_constraint=None,
    **kwargs
)
```

Important arguments:

Argument	Explanation
<i>filters</i>	Integer, the dimensionality of the output space (i.e. the number of output filters in the convolution).
<i>kernel_size</i>	An integer or tuple/list of 2 integers, specifying the height and width of the 2D convolution window. Can be a single integer to specify the same value for all spatial dimensions.
<i>strides</i>	integer or tuple/list of 2 integers, specifying the strides of the convolution along the height and width. Can be a single integer to specify the same value for all spatial dimensions. Specifying any stride value != 1 is incompatible with specifying any dilation_rate value != 1.
<i>padding</i>	one of "valid" or "same" (case-insensitive). "valid" means no padding. "same" results in padding with zeros evenly to the left/right or up/down of the input. When padding="same" and strides=1, the output has the same size as the input.

9.2. MaxPool2D

Max pooling operation for 2D spatial data.

The Max-Pooling operation requires selecting a kernel size and a stride length during architecture design. Once selected, the operation slides the kernel with the specified stride over the input while only selecting the largest value at each kernel slice from the input to yield a value for the output.

```
tf.keras.layers.  
    MaxPool2D( pool_size  
    = (2, 2), strides  
    =None , padding=  
    'valid', data_format  
    =None ,  
    ** kwargs
```


)

Important arguments:

Argument	Explanation
<i>pool_size</i>	integer or tuple of 2 integers, window size over which to take the maximum. (2, 2) will take the max value over a 2x2 pooling window. If only one integer is specified, the same window length will be used for both dimensions.
<i>strides</i>	integer, tuple of 2 integers, or None. Strides values. Specifies how far the pooling window moves for each pooling step. If None, it will default to pool_size.
<i>padding</i>	One of "valid" or "same" (case-insensitive). "valid" means no padding. "same" results in padding evenly to the left/right or up/down of the input such that output has the same height/width dimension as the input.

Example

```
x = tf.constant([[1., 2., 3.],
                 [4., 5., 6.],
                 [7., 8., 9.]])
x = tf.reshape(x, [1, 3, 3, 1])
max_pool_2d = tf.keras.layers.MaxPooling2D(pool_size=(2,
2), strides=(1, 1), padding='valid') max_pool_2d(x)
```

Output:

```
<tf.Tensor: shape=(1, 2, 2, 1), dtype=float32, numpy=
array([[[[5.],
         [6.]],
        [[8.],
         [9.]]]], dtype=float32)>
```

9.3. Flatten

Flattens the input. Does not affect the batch size.

This layer converts a three-dimensional layer in the network into a one-dimensional vector to fit the input of a fully-connected layer for classification. For example, a 5x5x2 tensor would be converted into a vector of size 50. The previous convolutional layers of the network extracted the features from the input image, but now it is time to classify the features. We use the softmax function to classify these features, which requires a 1-dimensional input. This is why the flatten layer is necessary.

```
tf . keras . layers . Flatten(  
    data_format =None ,    ** kwargs  
)
```

```
Example model = tf. keras. Sequential() model.  
add(tf . keras . layers. Conv2D(64, 3, 3, input_shape=  
(3,  
    32, 32)))  
print(model. output_shape)
```

Output:

```
(None, 1, 10, 64)
```

```
model . add(Flatten())  
model . output_shape
```

Output:

```
(None, 640)
```

9.4. Dense

Just your regular densely-connected NN layer.

Dense layer is the regular deeply connected neural network layer. It is most common and frequently used layer. Dense layer does the below operation on the input and returns the output.

```
tf.keras.layers.Dense(  
    units, activation=None,  
    use_bias=True,  
    kernel_initializer=  
    'glorot_uniform', bias_initializer  
    = 'zeros', kernel_regularizer  
    =None, bias_regularizer=None,  
    activity_regularizer=None,  
    kernel_constraint=None,  
    bias_constraint=None,  
    **kwargs  
)
```

Dense implements the operation: $\text{output} = \text{activation}(\text{dot}(\text{input}, \text{kernel}) + \text{bias})$ where activation is the element-wise activation function passed as the

activation argument, kernel is a weights matrix created by the layer, and bias is a bias vector created by the layer (only applicable if use_bias is True). These are all attributes of Dense.

Important arguments

Argument	Explanation
<i>units</i>	Positive integer, dimensionality of the output space.
<i>activation</i>	Activation function to use. If you don't specify anything, no activation is applied (ie. "linear" activation: $a(x) = x$).

10. BIRD vs DRONE IMAGE CLASSIFICATION

10.1. Import required dependencies and modules

```
import tensorflow as tf
import numpy as np import
pandas as pd import
matplotlib.pyplot as plt

from tensorflow.keras.preprocessing.image import
ImageDataGenerator from tensorflow.keras.optimizers import Adam
from tensorflow.keras.layers import Dense, Flatten,
Conv2D, MaxPool2D, Activation, BatchNormalization from
tensorflow.keras import Sequential from
tensorflow.keras.callbacks import ModelCheckpoint from
tensorflow.keras.models import load_model

import random
import os
import shutil
import zipfile
import pathlib
```

10.2. Mount Google Drive

```
from google.colab import drive

drive.mount( '/content/drive' )

Mounted at /content/drive
```

10.3 Download Dataset from Kaggle

```
! pip install kaggle

Looking in indexes: https://pypi.org/simple, https://us-
python.pkg.dev/colab-wheels/public/simple/
Requirement already satisfied: kaggle in
/usr/local/lib/python3.7/dist-packages (1.5.12)
```

```
Requirement already satisfied: requests in
/usr/local/lib/python3.7/dist-packages (from kaggle) (2.23.0)
Requirement already satisfied: certifi in
/usr/local/lib/python3.7/dist-packages (from kaggle) (2022.5.18.1)
Requirement already satisfied: six>=1.10 in
/usr/local/lib/python3.7/dist-packages (from kaggle) (1.15.0)
Requirement already satisfied: python-dateutil in
/usr/local/lib/python3.7/dist-packages (from kaggle) (2.8.2)
Requirement already satisfied: python-slugify in
/usr/local/lib/python3.7/dist-packages (from kaggle) (6.1.2)
Requirement already satisfied: tqdm in
/usr/local/lib/python3.7/dist-packages (from kaggle) (4.64.0)
Requirement already satisfied: urllib3 in
/usr/local/lib/python3.7/dist-packages (from kaggle) (1.24.3)
Requirement already satisfied: text-unidecode>=1.3 in
/usr/local/lib/python3.7/dist-packages (from python-slugify-
>kaggle) (1.3)
Requirement already satisfied: idna<3,>=2.5 in
/usr/local/lib/python3.7/dist-packages (from requests->kaggle)
(2.10)
Requirement already satisfied: chardet<4,>=3.0.2 in
/usr/local/lib/python3.7/dist-packages (from requests->kaggle)
(3.0.4)
```

```
source = "/content/drive/MyDrive/kaggle.json"
destination = "/content/"
shutil.copy(src=source, dst=destination)
'/content/kaggle.json'
```

```
! mkdir ~/.kaggle
! cp kaggle.json ~/.kaggle/
! chmod 600 ~/.kaggle/kaggle.json
! kaggle datasets download harshwalia/birds-vs-drone-
dataset
```

```
Downloading birds-vs-drone-dataset.zip to /content
85% 66.0M/78.1M [00:00<00:00, 124MB/s]
100% 78.1M/78.1M [00:00<00:00, 106MB/s]
```

```
# Unzip the downloaded file zip_ref =
zipfile.ZipFile( "birds-vs-drone-dataset.zip" )
zip_ref.extractall() zip_ref.close()
```

10.4 Split the dataset into training and test sets

```
def split_data ( SOURCE, TRAINING, TESTING, SPLIT_SIZE ):
    file_names = os.listdir(SOURCE)          random_file_names =
    random.sample(file_names, len (file_names)) split_threshold
    = round (SPLIT_SIZE * len (random_file_names)) for item
    in random_file_names[:split_threshold]:
        if os.path.getsize(os.path.join(SOURCE,item)) == 0 :
            print( f"{item} is zero length, so ignoring." )
        else:
            shutil.copy(os.path.join(SOURCE,item), TRAINING)
    for item in random_file_names[split_threshold :
len (random_file_names)]: if
    os.path.getsize(os.path.join(SOURCE,item)) == 0
    :
        print ( f"{item} is zero length, so ignoring."
    ) else :
        shutil.copy(os.path.join(SOURCE,item), TESTING)

DRONE_SOURCE_DIR = "/content/BirdVsDrone/Drones"
BIRD_SOURCE_DIR = "/content/BirdVsDrone/Birds"
TRAINING_DIR = '/content/BirdVsDroneSplit/train/'
TESTING_DIR = '/content/BirdVsDroneSplit/test/'
TRAINING_DRONE_DIR = TRAINING_DIR + "drones/"
TESTING_DRONE_DIR = TESTING_DIR + "drones/"
TRAINING_BIRD_DIR = TRAINING_DIR + "birds/"
TESTING_BIRD_DIR = TESTING_DIR + "birds/"

try :
    os.makedirs(TRAINING_BIRD_DIR)
os.makedirs(TESTING_BIRD_DIR)
os.makedirs(TRAINING_DRONE_DIR)
os.makedirs(TESTING_DRONE_DIR)
except : Pass
```

```
split_size = 0.8 split_data(DRONE_SOURCE_DIR,
TRAINING_DRONE_DIR, TESTING_DRONE_DIR, split_size)
split_data(BIRD_SOURCE_DIR, TRAINING_BIRD_DIR, TESTING_BIRD_DIR,
split_size)
```

10.5. Load the split dataset from Google Drive

```
dataset_path = "/content/drive/MyDrive/BirdVsDroneSplit"
for dirpath, dirnames, filenames in
os.walk(dataset_path):
    print (f"There are { len (dirnames)} directories and
{ len (filenames)} images in '{dirpath}'." )
```

There are 2 directories and 0 images in
'/content/drive/MyDrive/BirdVsDroneSplit'.

There are 2 directories and 0 images in
'/content/drive/MyDrive/BirdVsDroneSplit/test'.

There are 0 directories and 80 images in
'/content/drive/MyDrive/BirdVsDroneSplit/test/birds'.

There are 0 directories and 86 images in
'/content/drive/MyDrive/BirdVsDroneSplit/test/drones'.

There are 2 directories and 0 images in
'/content/drive/MyDrive/BirdVsDroneSplit/train'.

There are 0 directories and 342 images in
'/content/drive/MyDrive/BirdVsDroneSplit/train/drones'.

There are 0 directories and 320 images in
'/content/drive/MyDrive/BirdVsDroneSplit/train/birds'.

```
# Get the class names programmatically data_dir =
pathlib.Path(dataset_path + "/train" ) class_names =
np.array( sorted ([item.name for item in data_dir.glob(
"*" )])) # Created a list of class_names from the
subdirectories print (class_names)
['birds', 'drones']
```


10.6. Visualize random images from dataset

```
# Let's visualize our images import
matplotlib.image as mpimg def
view_random_image( target_dir, target_class ):
    # Setup the target directory (we'll view images from here)
    target_folder = target_dir + target_class
    # Get a random image path random_image =
    random.sample(os.listdir(target_folder), 1 ) print
    (random_image)
    # Read in the image and plot it using matplotlib img =
    mpimg.imread(target_folder + "/" + random_image[0 ])
    plt.imshow(img) plt.title(target_class) plt.axis( "off"
    ); print ( f"Image shape: {img.shape}" ) # show the
    shape of the image return img
```

```
# View a random image from the training dataset img =
view_random_image(target_dir=dataset_path + "/train/"
,
target_class= "birds" )
['singleBirdinsky164.jpeg']
Image shape: (168, 299, 3)
```



```
# View a random image from the training dataset img =
view_random_image(target_dir=dataset_path + "/train/" ,
target_class= "drones" )
['singleDronesinsky13.jpeg']
```

Image shape: (410, 615, 3)



10.7. Prepare the dataset to feed into the neural network

```
train_datagen = ImageDataGenerator(rescale= 1 / 255.0 )
test_datagen = ImageDataGenerator(rescale= 1/ 255.0 )

train_dir = dataset_path +  "/train/"
test_dir = dataset_path +  "/test/"

train_data = train_datagen.flow_from_directory(
    directory=train_dir,          # Target directory of images
    target_size=( 224 ,  224 ),  # Target size of images (height,
width)  class_mode="binary",    # Type of data you're working
    with  batch_size=32 ,        # Size of minibatches to load
    data
into  shuffle=
    True )

test_data = test_datagen.flow_from_directory(directory=test_dir,
```

```
target_size=( 224 ,
224 ),
class_mode="binary",
batch_size= 32 ,
shuffle= True)
```

Found 660 images belonging to 2 classes.
Found 166 images belonging to 2 classes.

```
train_datagen_augmented = ImageDataGenerator(rescale= 1/ 255. ,
rotation_range= 0.2 , # how much do you want to rotate an
image?
shear_range= 0.2 , # how much do you want to shear
an
image?
zoom_range= 0.2 , # zoom in randomly on an
image
width_shift_range= 0.2 , # move your image around on
the
x-axis
height_shift_range= 0.2 # move your image around
on the
y-axis
horizontal_flip= True ) # do you want to flip an
image?

train_data_augmented =
train_datagen_augmented.flow_from_directory(directory=train_dir,
target_size=(224 , 224 ),class_mode="binary" ,batch_size=32 )
Found 660 images belonging to 2 classes.

saved_models_path = "/content/drive/MyDrive/saved_models/"
```

10.8 Define helper functions

```
# Plot the validation and training curves separately
def plot_loss_curves ( history ):
```

```

    """ Returns separate loss curves for training and validation
    metrics
    """
    loss = history.history[ "loss" ]
    val_loss = history.history[ "val_loss" ]
    accuracy = history.history[ "accuracy" ]
    val_accuracy =
    history.history["val_accuracy" ]

    epochs = range ( len (history.history[ "loss" ])) # how
                                                         many
    epochs did we run for? # Plot loss plt.plot(epochs,
    loss, label= "training_loss" ) plt.plot(epochs,
    val_loss, label= "val_loss") plt.title( "Loss" )
    plt.xlabel( "Epochs" ) plt.legend() # Plot
    accuracy plt.figure() plt.plot(epochs,
    accuracy, label= "training_accuracy" )
    plt.plot(epochs, val_accuracy, label=
    "val_accuracy") plt.title("Accuracy" ) plt.xlabel(
    "Epochs") plt.legend()

# Create a function to import an image and resize it to be able
to be used with our model
def load_and_prep_image ( filename,
img_shape=224 ):
    """
    Reads an image from filename, turns it into a tensor
    and reshapes it to (img_shape, img_shape,
    colour_channels).
    """
    # Read in the image img =
    tf.io.read_file(filename) # Decode
    the read file into a tensor img
    = tf.image.decode_image(img)
    # Resize the image img = tf.image.resize(img,
    size=[img_shape, img_shape]) # Rescale the image
    (get all values between 0 and 1) img = img / 255.
    return img

def pred_and_plot ( model, filename, class_names=class_names ):

```

```

"""
Imports an image located at filename, makes a prediction with
model and plots the image with the predicted class as the title.
"""

# Import the target image and preprocess it
img = load_and_prep_image(filename)

# Make a prediction    pred =
model.predict(tf.expand_dims(img, axis= 0 ))

print      (pred)

# Get the predicted class    pred_class =
class_names[ int (tf. round (pred))] # Plot
the image and predicted class
plt.imshow(img)    plt.title( f"Prediction:
{pred_class}")    plt.axis(      False );

def  pred_and_plot_directory ( model, directory,
class_names=class_names ):
    """
Imports an image located at directory, makes a prediction with
model and plots the image with the predicted class as the title.
"""

# Import the target image and preprocess
it      plt.figure(figsize=(12 , 12 ))    i
= 1    for  filename in
os.listdir(directory):
    img = load_and_prep_image(directory + "/" + filename)
    # Make a prediction    pred =
    model.predict(tf.expand_dims(img, axis= 0 ))
    prob = pred[ 0 ][ 0 ]
    # print(pred)
    # Get the predicted class
    pred_class = class_names[ int (tf. round (pred))]
    # Calculate probability if pred_class is bird
    if  pred_class ==  'birds' :
        prob =  1  - prob
    # Plot the image and predicted class
    plt.subplot(2 , 2 , i)    plt.imshow(img)
    plt.title(  "Prediction: {} \nProbability:
{: .4f}"

```

```

. format (pred_class,
          prob)) plt.axis(
          False      ); i
          +=          1

def walk_through_dir ( dir_path
): """
Walks through dir_path returning its contents.
Args: dir_path (str): target
directory      Returns:
A print out of:  number of subdirectories in
dir_path number of images (files) in each
subdirectory  name of each subdirectory
""" for dirpath, dirnames, filenames in
os.walk(dir_path):
    print ( f"There are { len (dirnames)} directories and
{ len (filenames)} images in '{dirpath}'." )

```

10.9. Create the model, Compile the model and Fit the model

```

tf.random.set_seed( 42 )

model_4 = Sequential([
    # Note the input shape is the desired size of the image 224 *
    224  with 3 bytes color
    # This is the first convolution
    Conv2D( 16 , ( 3 , 3 ), activation= 'relu' , input_shape=( 224 ,
    224 , 3 )),
    Conv2D( 16 , ( 3 , 3 ), activation= 'relu' ),
    MaxPool2D( 2 , 2 ),
    # The second convolution
    Conv2D( 32 , (4 , 4 ), activation= 'relu'
), Conv2D( 32 , ( 4, 4 ), activation=
'relu' ),
    MaxPool2D( 2 ,2 ),

```

```

    # The third convolution
    Conv2D( 64 , (5 , 5 ) , activation= 'relu'
),  Conv2D( 64 , (5 , 5 ) , activation=
'relu' ) ,
    MaxPool2D(2 , 2 ) ,

    # Flatten the results to feed into a DNN
    Flatten() ,

    # 512 neuron hidden layer
    Dense( 256 , activation= 'relu' ) ,
    Dense( 512 , activation= 'relu' ) ,

    # Only 1 output neuron. It will contain a value from 0-1 where 0
for 1 class ('drones') and 1 for the other ('birds')
    Dense( 1 , activation= 'sigmoid' )
] )

model_4. compile (loss= "binary_crossentropy" ,
                  optimizer=Adam() ,
                  metrics=[ "accuracy" ])

callback = ModelCheckpoint(filepath=saved_models_path +  "/model_4"
,

                           monitor= "val_accuracy"
                           ,  save_best_only= True ,
                           mode= "max" )

history_4 = model_4.fit(train_data_augmented,
                        epochs= 150 ,  steps_per_epoch= len
(train_data_augmented) ,
                        validation_data=test_data ,
                        validation_steps= len (test_data) ,
                        callbacks=[callback])

```

plot_loss_curves(history_4)

Epoch 1/150

21/21 [=====] - ETA: 0s - loss: 0.8064 - accuracy: 0.5242
WARNING:absl:Found untraced functions such as _jit_compiled_convolution_op, _jit_compiled_convolution_op, _jit_compiled_convolution_op, _jit_compiled_convolution_op while saving (showing 5 of 6). These functions will not be directly callable after loading.

21/21 [=====] - 20s 933ms/step - loss: 0.8064 - accuracy: 0.5242 - val_loss: 0.6823 - val_accuracy: 0.5181

Epoch 2/150

21/21 [=====] - ETA: 0s - loss: 0.6425 - accuracy: 0.6167
WARNING:absl:Found untraced functions such as _jit_compiled_convolution_op, _jit_compiled_convolution_op, _jit_compiled_convolution_op, _jit_compiled_convolution_op, _jit_compiled_convolution_op while saving (showing 5 of 6). These functions will not be directly callable after loading.

21/21 [=====] - 18s 858ms/step - loss: 0.6425 - accuracy: 0.6167 - val_loss: 0.6274 - val_accuracy: 0.6265

Epoch 3/150

21/21 [=====] - ETA: 0s - loss: 0.6028 - accuracy: 0.6636
WARNING:absl:Found untraced functions such as _jit_compiled_convolution_op, _jit_compiled_convolution_op, _jit_compiled_convolution_op, _jit_compiled_convolution_op, _jit_compiled_convolution_op while saving (showing 5 of 6). These functions will not be directly callable after loading.

21/21 [=====] - 18s 855ms/step - loss: 0.6028 - accuracy: 0.6636 - val_loss: 0.5134 - val_accuracy: 0.7410

Epoch 4/150

21/21 [=====] - 16s 780ms/step - loss: 0.5460 - accuracy: 0.7197 - val_loss: 0.5017 - val_accuracy: 0.7169

Epoch 5/150

21/21 [=====] - 16s 766ms/step - loss: 0.5241 - accuracy: 0.7273 - val_loss: 0.5151 - val_accuracy: 0.7289

Epoch 6/150

21/21 [=====] - ETA: 0s - loss: 0.5160 - accuracy: 0.7591
WARNING:absl:Found untraced functions such as _jit_compiled_convolution_op, _jit_compiled_convolution_op,


```

_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op while saving (showing 5 of 6). These
functions will not be directly callable after loading.
21/21 [=====] - 18s 869ms/step - loss:
0.5160 - accuracy: 0.7591 - val_loss: 0.4935 - val_accuracy: 0.7590
Epoch 7/150
21/21 [=====] - 17s 799ms/step - loss:
0.4944 - accuracy: 0.7727 - val_loss: 0.5602 - val_accuracy: 0.7108
Epoch 8/150
21/21 [=====] - ETA: 0s - loss: 0.5468 -
accuracy: 0.7424WARNING:absl:Found untraced functions such as
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op while saving (showing 5 of 6). These
functions will not be directly callable after loading.
21/21 [=====] - 18s 865ms/step - loss:
0.5468 - accuracy: 0.7424 - val_loss: 0.4648 - val_accuracy: 0.8012
Epoch 9/150
21/21 [=====] - ETA: 0s - loss: 0.4777 -
accuracy: 0.7879WARNING:absl:Found untraced functions such as
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op while saving (showing 5 of 6). These
functions will not be directly callable after loading.
21/21 [=====] - 18s 852ms/step - loss:
0.4777 - accuracy: 0.7879 - val_loss: 0.4478 - val_accuracy: 0.8072
Epoch 10/150
21/21 [=====] - 16s 791ms/step - loss:
0.4356 - accuracy: 0.8030 - val_loss: 0.7006 - val_accuracy: 0.6988
Epoch 11/150
21/21 [=====] - 16s 769ms/step - loss:
0.4872 - accuracy: 0.7818 - val_loss: 0.5108 - val_accuracy: 0.8072
Epoch 12/150
21/21 [=====] - 16s 780ms/step - loss:
0.4822 - accuracy: 0.7803 - val_loss: 0.4453 - val_accuracy: 0.7771
Epoch 13/150
21/21 [=====] - ETA: 0s - loss: 0.4853 -
accuracy: 0.7788WARNING:absl:Found untraced functions such as
_jit_compiled_convolution_op, _jit_compiled_convolution_op,

```

```

    _jit_compiled_convolution_op, _jit_compiled_convolution_op,
    _jit_compiled_convolution_op while saving (showing 5 of 6). These
    functions will not be directly callable after loading.
21/21 [=====] - 17s 844ms/step - loss:
0.4853 - accuracy: 0.7788 - val_loss: 0.4218 - val_accuracy: 0.8373
Epoch 14/150
21/21 [=====] - 16s 762ms/step - loss:
0.4341 - accuracy: 0.8288 - val_loss: 0.4318 - val_accuracy: 0.7892
Epoch 15/150
21/21 [=====] - 16s 768ms/step - loss:
0.4157 - accuracy: 0.8182 - val_loss: 0.4756 - val_accuracy: 0.7952
Epoch 16/150
21/21 [=====] - ETA: 0s - loss: 0.4413 -
accuracy: 0.8136WARNING:absl:Found untraced functions such as
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op while saving (showing 5 of 6). These
functions will not be directly callable after loading.
21/21 [=====] - 18s 833ms/step - loss:
0.4413 - accuracy: 0.8136 - val_loss: 0.3941 - val_accuracy: 0.8434
Epoch 17/150
21/21 [=====] - 16s 784ms/step - loss:
0.4002 - accuracy: 0.8273 - val_loss: 0.3918 - val_accuracy: 0.8373
Epoch 18/150
21/21 [=====] - 16s 750ms/step - loss:
0.4500 - accuracy: 0.8106 - val_loss: 0.4164 - val_accuracy: 0.8313
Epoch 19/150
21/21 [=====] - 16s 770ms/step - loss:
0.4684 - accuracy: 0.8045 - val_loss: 0.4183 - val_accuracy: 0.8133
Epoch 20/150
21/21 [=====] - 16s 752ms/step - loss:
0.4580 - accuracy: 0.8030 - val_loss: 0.5493 - val_accuracy: 0.7470
Epoch 21/150
21/21 [=====] - 16s 770ms/step - loss:
0.4448 - accuracy: 0.8152 - val_loss: 0.4168 - val_accuracy: 0.8313
Epoch 22/150
21/21 [=====] - 16s 764ms/step - loss:
0.4128 - accuracy: 0.8348 - val_loss: 0.4036 - val_accuracy: 0.8434
Epoch 23/150
21/21 [=====] - 16s 773ms/step - loss:

```

```

0.3902 - accuracy: 0.8212 - val_loss: 0.5771 - val_accuracy: 0.8012
Epoch 24/150
21/21 [=====] - 16s 751ms/step - loss:
0.3979 - accuracy: 0.8288 - val_loss: 0.3892 - val_accuracy: 0.8434
Epoch 25/150
21/21 [=====] - 16s 762ms/step - loss:
0.4089 - accuracy: 0.8212 - val_loss: 0.3944 - val_accuracy: 0.8313
Epoch 26/150
21/21 [=====] - 16s 754ms/step - loss:
0.3798 - accuracy: 0.8515 - val_loss: 0.4263 - val_accuracy: 0.8133
Epoch 27/150
21/21 [=====] - ETA: 0s - loss: 0.3893 -
accuracy: 0.8409WARNING:absl:Found untraced functions such as
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op while saving (showing 5 of 6). These
functions will not be directly callable after loading.
21/21 [=====] - 18s 850ms/step - loss:
0.3893 - accuracy: 0.8409 - val_loss: 0.3824 - val_accuracy: 0.8614
Epoch 28/150
21/21 [=====] - 16s 786ms/step - loss:
0.3681 - accuracy: 0.8545 - val_loss: 0.4229 - val_accuracy: 0.8072
Epoch 29/150
21/21 [=====] - ETA: 0s - loss: 0.3673 -
accuracy: 0.8485WARNING:absl:Found untraced functions such as
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op while saving (showing 5 of 6). These
functions will not be directly callable after loading.
21/21 [=====] - 18s 842ms/step - loss:
0.3673 - accuracy: 0.8485 - val_loss: 0.3410 - val_accuracy: 0.8735
Epoch 30/150
21/21 [=====] - 16s 784ms/step - loss:
0.3649 - accuracy: 0.8515 - val_loss: 0.4204 - val_accuracy: 0.8133
Epoch 31/150
21/21 [=====] - ETA: 0s - loss: 0.3531 -
accuracy: 0.8409WARNING:absl:Found untraced functions such as
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op while saving (showing 5 of 6). These
functions will not be directly callable after loading.

```

21/21 [=====] - 17s 831ms/step - loss:
0.3531 - accuracy: 0.8409 - val_loss: 0.3238 - val_accuracy: 0.8916
Epoch 32/150

21/21 [=====] - 16s 775ms/step - loss:
0.3097 - accuracy: 0.8712 - val_loss: 0.3281 - val_accuracy: 0.8675
Epoch 33/150

21/21 [=====] - 16s 764ms/step - loss:
0.3019 - accuracy: 0.8667 - val_loss: 0.3030 - val_accuracy: 0.8855
Epoch 34/150

21/21 [=====] - 16s 769ms/step - loss:
0.3312 - accuracy: 0.8697 - val_loss: 0.3303 - val_accuracy: 0.8614
Epoch 35/150

21/21 [=====] - 16s 771ms/step - loss:
0.3194 - accuracy: 0.8591 - val_loss: 0.3026 - val_accuracy: 0.8675
Epoch 36/150

21/21 [=====] - 16s 767ms/step - loss:
0.3151 - accuracy: 0.8606 - val_loss: 0.3985 - val_accuracy: 0.8193
Epoch 37/150

21/21 [=====] - 16s 758ms/step - loss:
0.3100 - accuracy: 0.8697 - val_loss: 0.3569 - val_accuracy: 0.8554
Epoch 38/150

21/21 [=====] - 16s 758ms/step - loss:
0.3449 - accuracy: 0.8409 - val_loss: 0.5394 - val_accuracy: 0.7651
Epoch 39/150

21/21 [=====] - 16s 763ms/step - loss:
0.3418 - accuracy: 0.8606 - val_loss: 0.3550 - val_accuracy: 0.8614
Epoch 40/150

21/21 [=====] - 16s 757ms/step - loss:
0.3401 - accuracy: 0.8545 - val_loss: 0.3549 - val_accuracy: 0.8855
Epoch 41/150

21/21 [=====] - 16s 753ms/step - loss:
0.3033 - accuracy: 0.8818 - val_loss: 0.3415 - val_accuracy: 0.8795
Epoch 42/150

21/21 [=====] - 16s 765ms/step - loss:
0.2908 - accuracy: 0.8803 - val_loss: 0.3087 - val_accuracy: 0.8675
Epoch 43/150

21/21 [=====] - 16s 769ms/step - loss:
0.3284 - accuracy: 0.8758 - val_loss: 0.3703 - val_accuracy: 0.8675
Epoch 44/150

```

21/21 [=====] - 16s 776ms/step - loss:
0.3043 - accuracy: 0.8758 - val_loss: 0.4295 - val_accuracy: 0.8434
Epoch 45/150
21/21 [=====] - 16s 780ms/step - loss:
0.3246 - accuracy: 0.8652 - val_loss: 0.3901 - val_accuracy: 0.8614
Epoch 46/150
21/21 [=====] - 16s 788ms/step - loss:
0.2897 - accuracy: 0.8788 - val_loss: 0.5113 - val_accuracy: 0.8675
Epoch 47/150
21/21 [=====] - 16s 782ms/step - loss:
0.3554 - accuracy: 0.8697 - val_loss: 0.3796 - val_accuracy: 0.8795
Epoch 48/150
21/21 [=====] - 16s 765ms/step - loss:
0.3317 - accuracy: 0.8758 - val_loss: 0.3479 - val_accuracy: 0.8554
Epoch 49/150
21/21 [=====] - 16s 762ms/step - loss:
0.3174 - accuracy: 0.8576 - val_loss: 0.3628 - val_accuracy: 0.8554
Epoch 50/150
21/21 [=====] - 16s 770ms/step - loss:
0.3062 - accuracy: 0.8788 - val_loss: 0.2972 - val_accuracy: 0.8614
Epoch 51/150
21/21 [=====] - 16s 771ms/step - loss:
0.2823 - accuracy: 0.8833 - val_loss: 0.2833 - val_accuracy: 0.8675
Epoch 52/150
21/21 [=====] - 16s 754ms/step - loss:
0.2735 - accuracy: 0.8939 - val_loss: 0.3319 - val_accuracy: 0.8675
Epoch 53/150
21/21 [=====] - 16s 758ms/step - loss:
0.2728 - accuracy: 0.8939 - val_loss: 0.2955 - val_accuracy: 0.8795
Epoch 54/150
21/21 [=====] - ETA: 0s - loss: 0.2757 -
accuracy: 0.8848WARNING:absl:Found untraced functions such as
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op while saving (showing 5 of 6). These
functions will not be directly callable after loading.
21/21 [=====] - 17s 835ms/step - loss:
0.2757 - accuracy: 0.8848 - val_loss: 0.2846 - val_accuracy: 0.9036
Epoch 55/150
21/21 [=====] - 16s 782ms/step - loss:

```

0.2521 - accuracy: 0.9000 - val_loss: 0.2911 - val_accuracy: 0.8855
Epoch 56/150
21/21 [=====] - 16s 757ms/step - loss:
0.3060 - accuracy: 0.8621 - val_loss: 0.2652 - val_accuracy: 0.8795
Epoch 57/150
21/21 [=====] - 16s 750ms/step - loss:
0.2645 - accuracy: 0.8909 - val_loss: 0.2706 - val_accuracy: 0.8855
Epoch 58/150
21/21 [=====] - 16s 766ms/step - loss:
0.2598 - accuracy: 0.9136 - val_loss: 0.3138 - val_accuracy: 0.8494
Epoch 59/150
21/21 [=====] - 16s 764ms/step - loss:
0.2405 - accuracy: 0.9030 - val_loss: 0.2751 - val_accuracy: 0.8735
Epoch 60/150
21/21 [=====] - 16s 760ms/step - loss:
0.2194 - accuracy: 0.9121 - val_loss: 0.2777 - val_accuracy: 0.8976
Epoch 61/150
21/21 [=====] - 16s 759ms/step - loss:
0.2111 - accuracy: 0.9167 - val_loss: 0.3424 - val_accuracy: 0.8916
Epoch 62/150
21/21 [=====] - 16s 760ms/step - loss:
0.2648 - accuracy: 0.8924 - val_loss: 0.3164 - val_accuracy: 0.8494
Epoch 63/150
21/21 [=====] - 16s 745ms/step - loss:
0.2182 - accuracy: 0.9045 - val_loss: 0.2838 - val_accuracy: 0.8916
Epoch 64/150
21/21 [=====] - 16s 754ms/step - loss:
0.2783 - accuracy: 0.8818 - val_loss: 0.3350 - val_accuracy: 0.8494
Epoch 65/150
21/21 [=====] - 16s 754ms/step - loss:
0.2689 - accuracy: 0.8788 - val_loss: 0.2551 - val_accuracy: 0.8795
Epoch 66/150
21/21 [=====] - 16s 762ms/step - loss:
0.2436 - accuracy: 0.9091 - val_loss: 0.2753 - val_accuracy: 0.8916
Epoch 67/150
21/21 [=====] - 16s 771ms/step - loss:
0.2511 - accuracy: 0.9030 - val_loss: 0.2997 - val_accuracy: 0.8855
Epoch 68/150
21/21 [=====] - 16s 756ms/step - loss:

```

0.2210 - accuracy: 0.9091 - val_loss: 0.2753 - val_accuracy: 0.8795
Epoch 69/150
21/21 [=====] - 16s 772ms/step - loss:
0.2136 - accuracy: 0.9015 - val_loss: 0.2774 - val_accuracy: 0.8976
Epoch 70/150
21/21 [=====] - 16s 752ms/step - loss:
0.2144 - accuracy: 0.9091 - val_loss: 0.2480 - val_accuracy: 0.8976
Epoch 71/150
21/21 [=====] - ETA: 0s - loss: 0.1836 -
accuracy: 0.9197WARNING:absl:Found untraced functions such as
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op while saving (showing 5 of 6). These
functions will not be directly callable after loading.
21/21 [=====] - 18s 859ms/step - loss:
0.1836 - accuracy: 0.9197 - val_loss: 0.2783 - val_accuracy: 0.9337
Epoch 72/150
21/21 [=====] - 16s 780ms/step - loss:
0.2256 - accuracy: 0.8985 - val_loss: 0.5752 - val_accuracy: 0.8976
Epoch 73/150
21/21 [=====] - 16s 773ms/step - loss:
0.2919 - accuracy: 0.8985 - val_loss: 0.2905 - val_accuracy: 0.9036
Epoch 74/150
21/21 [=====] - 16s 777ms/step - loss:
0.1984 - accuracy: 0.9227 - val_loss: 0.2682 - val_accuracy: 0.8916
Epoch 75/150
21/21 [=====] - 16s 775ms/step - loss:
0.2034 - accuracy: 0.9106 - val_loss: 0.2556 - val_accuracy: 0.9217
Epoch 76/150
21/21 [=====] - 16s 768ms/step - loss:
0.2010 - accuracy: 0.9273 - val_loss: 0.3289 - val_accuracy: 0.8976
Epoch 77/150
21/21 [=====] - 16s 763ms/step - loss:
0.2241 - accuracy: 0.9045 - val_loss: 0.3303 - val_accuracy: 0.8916
Epoch 78/150
21/21 [=====] - 16s 745ms/step - loss:
0.1979 - accuracy: 0.9136 - val_loss: 0.3917 - val_accuracy: 0.8614
Epoch 79/150
21/21 [=====] - 16s 753ms/step - loss:
0.1995 - accuracy: 0.9197 - val_loss: 0.2808 - val_accuracy: 0.8735

```

```

Epoch 80/150
21/21 [=====] - 16s 764ms/step - loss:
0.2089 - accuracy: 0.9091 - val_loss: 0.2934 - val_accuracy: 0.8855
Epoch 81/150
21/21 [=====] - 16s 754ms/step - loss:
0.1843 - accuracy: 0.9091 - val_loss: 0.3291 - val_accuracy: 0.9217
Epoch 82/150
21/21 [=====] - 16s 754ms/step - loss:
0.1931 - accuracy: 0.9258 - val_loss: 0.2849 - val_accuracy: 0.9157
Epoch 83/150
21/21 [=====] - 16s 757ms/step - loss:
0.1997 - accuracy: 0.9106 - val_loss: 0.2449 - val_accuracy: 0.9217
Epoch 84/150
21/21 [=====] - 16s 767ms/step - loss:
0.1625 - accuracy: 0.9258 - val_loss: 0.2389 - val_accuracy: 0.9157
Epoch 85/150
21/21 [=====] - 16s 765ms/step - loss:
0.1468 - accuracy: 0.9348 - val_loss: 0.1919 - val_accuracy: 0.9277
Epoch 86/150
21/21 [=====] - 16s 771ms/step - loss:
0.1979 - accuracy: 0.9273 - val_loss: 0.3289 - val_accuracy: 0.8313
Epoch 87/150
21/21 [=====] - 16s 769ms/step - loss:
0.2078 - accuracy: 0.9000 - val_loss: 0.3014 - val_accuracy: 0.8735
Epoch 88/150
21/21 [=====] - ETA: 0s - loss: 0.1931 -
accuracy: 0.9106WARNING:absl:Found untraced functions such as
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op while saving (showing 5 of 6). These
functions will not be directly callable after loading.
21/21 [=====] - 18s 845ms/step - loss:
0.1931 - accuracy: 0.9106 - val_loss: 0.1754 - val_accuracy: 0.9458
Epoch 89/150
21/21 [=====] - 16s 779ms/step - loss:
0.1520 - accuracy: 0.9333 - val_loss: 0.3259 - val_accuracy: 0.8675
Epoch 90/150
21/21 [=====] - 16s 761ms/step - loss:
0.2031 - accuracy: 0.9227 - val_loss: 0.2233 - val_accuracy: 0.9277

```


Epoch 91/150
21/21 [=====] - 16s 755ms/step - loss:
0.1498 - accuracy: 0.9379 - val_loss: 0.3125 - val_accuracy: 0.9157
Epoch 92/150
21/21 [=====] - 16s 746ms/step - loss:
0.1571 - accuracy: 0.9470 - val_loss: 0.3551 - val_accuracy: 0.8735
Epoch 93/150
21/21 [=====] - 16s 753ms/step - loss:
0.1686 - accuracy: 0.9288 - val_loss: 0.3368 - val_accuracy: 0.9157
Epoch 94/150
21/21 [=====] - 16s 757ms/step - loss:
0.1538 - accuracy: 0.9273 - val_loss: 0.4729 - val_accuracy: 0.8795
Epoch 95/150
21/21 [=====] - 16s 753ms/step - loss:
0.3101 - accuracy: 0.8894 - val_loss: 0.4396 - val_accuracy: 0.8373
Epoch 96/150
21/21 [=====] - 16s 752ms/step - loss:
0.1937 - accuracy: 0.9061 - val_loss: 0.3373 - val_accuracy: 0.8976
Epoch 97/150
21/21 [=====] - 16s 761ms/step - loss:
0.1979 - accuracy: 0.9227 - val_loss: 0.2158 - val_accuracy: 0.9096
Epoch 98/150
21/21 [=====] - 16s 756ms/step - loss:
0.2195 - accuracy: 0.9136 - val_loss: 0.2726 - val_accuracy: 0.9036
Epoch 99/150
21/21 [=====] - 16s 756ms/step - loss:
0.2029 - accuracy: 0.9197 - val_loss: 0.1785 - val_accuracy: 0.9096
Epoch 100/150
21/21 [=====] - 16s 773ms/step - loss:
0.2155 - accuracy: 0.8970 - val_loss: 0.2190 - val_accuracy: 0.9217
Epoch 101/150
21/21 [=====] - 16s 748ms/step - loss:
0.2402 - accuracy: 0.9076 - val_loss: 0.2343 - val_accuracy: 0.9277
Epoch 102/150
21/21 [=====] - 16s 760ms/step - loss:
0.2132 - accuracy: 0.8985 - val_loss: 0.3032 - val_accuracy: 0.9157
Epoch 103/150
21/21 [=====] - 16s 759ms/step - loss:
0.1782 - accuracy: 0.9288 - val_loss: 0.3544 - val_accuracy: 0.9096

Epoch 104/150
21/21 [=====] - 16s 756ms/step - loss:
0.1753 - accuracy: 0.9242 - val_loss: 0.4839 - val_accuracy: 0.8735
Epoch 105/150
21/21 [=====] - 16s 749ms/step - loss:
0.1753 - accuracy: 0.9288 - val_loss: 0.4785 - val_accuracy: 0.8855
Epoch 106/150
21/21 [=====] - 16s 759ms/step - loss:
0.1955 - accuracy: 0.9106 - val_loss: 0.3506 - val_accuracy: 0.9337
Epoch 107/150
21/21 [=====] - 16s 761ms/step - loss:
0.1880 - accuracy: 0.9288 - val_loss: 0.4176 - val_accuracy: 0.8795
Epoch 108/150
21/21 [=====] - 16s 756ms/step - loss:
0.1679 - accuracy: 0.9227 - val_loss: 0.3679 - val_accuracy: 0.8976
Epoch 109/150
21/21 [=====] - 16s 757ms/step - loss:
0.1366 - accuracy: 0.9424 - val_loss: 0.3936 - val_accuracy: 0.9217
Epoch 110/150
21/21 [=====] - 16s 780ms/step - loss:
0.1507 - accuracy: 0.9364 - val_loss: 0.3405 - val_accuracy: 0.9157
Epoch 111/150
21/21 [=====] - 16s 760ms/step - loss:
0.2200 - accuracy: 0.9333 - val_loss: 0.3036 - val_accuracy: 0.8976
Epoch 112/150
21/21 [=====] - 16s 754ms/step - loss:
0.1946 - accuracy: 0.9242 - val_loss: 0.3604 - val_accuracy: 0.9096
Epoch 113/150
21/21 [=====] - 16s 770ms/step - loss:
0.1578 - accuracy: 0.9379 - val_loss: 0.4720 - val_accuracy: 0.9036
Epoch 114/150
21/21 [=====] - 16s 749ms/step - loss:
0.1621 - accuracy: 0.9288 - val_loss: 0.4016 - val_accuracy: 0.9096
Epoch 115/150
21/21 [=====] - 16s 762ms/step - loss:
0.1541 - accuracy: 0.9333 - val_loss: 1.4418 - val_accuracy: 0.8916
Epoch 116/150
21/21 [=====] - 16s 747ms/step - loss:
0.1739 - accuracy: 0.9242 - val_loss: 0.5817 - val_accuracy: 0.8434

Epoch 117/150
21/21 [=====] - 16s 755ms/step - loss:
0.2097 - accuracy: 0.9045 - val_loss: 0.4117 - val_accuracy: 0.8855
Epoch 118/150
21/21 [=====] - 16s 747ms/step - loss:
0.1615 - accuracy: 0.9288 - val_loss: 0.3728 - val_accuracy: 0.9458
Epoch 119/150
21/21 [=====] - 16s 755ms/step - loss:
0.1375 - accuracy: 0.9515 - val_loss: 0.3361 - val_accuracy: 0.9096
Epoch 120/150
21/21 [=====] - 16s 762ms/step - loss:
0.1675 - accuracy: 0.9409 - val_loss: 0.6814 - val_accuracy: 0.8434
Epoch 121/150
21/21 [=====] - 16s 750ms/step - loss:
0.2244 - accuracy: 0.9136 - val_loss: 0.4012 - val_accuracy: 0.8795
Epoch 122/150
21/21 [=====] - 16s 752ms/step - loss:
0.1459 - accuracy: 0.9470 - val_loss: 0.3541 - val_accuracy: 0.9458
Epoch 123/150
21/21 [=====] - 16s 751ms/step - loss:
0.1450 - accuracy: 0.9455 - val_loss: 0.4096 - val_accuracy: 0.9337
Epoch 124/150
21/21 [=====] - 16s 751ms/step - loss:
0.1151 - accuracy: 0.9561 - val_loss: 0.3552 - val_accuracy: 0.9337
Epoch 125/150
21/21 [=====] - 16s 755ms/step - loss:
0.1714 - accuracy: 0.9409 - val_loss: 0.4630 - val_accuracy: 0.9398
Epoch 126/150
21/21 [=====] - 16s 756ms/step - loss:
0.1312 - accuracy: 0.9515 - val_loss: 0.3732 - val_accuracy: 0.9337
Epoch 127/150
21/21 [=====] - 16s 744ms/step - loss:
0.1328 - accuracy: 0.9394 - val_loss: 0.3800 - val_accuracy: 0.9157
Epoch 128/150
21/21 [=====] - 16s 751ms/step - loss:
0.1579 - accuracy: 0.9273 - val_loss: 0.5696 - val_accuracy: 0.9398
Epoch 129/150
21/21 [=====] - 16s 747ms/step - loss:
0.1824 - accuracy: 0.9288 - val_loss: 0.5185 - val_accuracy: 0.8614

Epoch 130/150
21/21 [=====] - 16s 758ms/step - loss:
0.1811 - accuracy: 0.9212 - val_loss: 0.3545 - val_accuracy: 0.9458
Epoch 131/150
21/21 [=====] - 16s 760ms/step - loss:
0.1545 - accuracy: 0.9394 - val_loss: 0.2992 - val_accuracy: 0.8916
Epoch 132/150
21/21 [=====] - 16s 754ms/step - loss:
0.1591 - accuracy: 0.9379 - val_loss: 0.2749 - val_accuracy: 0.9337
Epoch 133/150
21/21 [=====] - 16s 748ms/step - loss:
0.1173 - accuracy: 0.9530 - val_loss: 0.3884 - val_accuracy: 0.8976
Epoch 134/150
21/21 [=====] - 16s 743ms/step - loss:
0.1084 - accuracy: 0.9576 - val_loss: 0.3139 - val_accuracy: 0.9337
Epoch 135/150
21/21 [=====] - 16s 759ms/step - loss:
0.0938 - accuracy: 0.9591 - val_loss: 0.3251 - val_accuracy: 0.9217
Epoch 136/150
21/21 [=====] - 16s 757ms/step - loss:
0.0967 - accuracy: 0.9591 - val_loss: 0.2684 - val_accuracy: 0.9398
Epoch 137/150
21/21 [=====] - 16s 757ms/step - loss:
0.1287 - accuracy: 0.9500 - val_loss: 0.2999 - val_accuracy: 0.9036
Epoch 138/150
21/21 [=====] - 16s 748ms/step - loss:
0.0966 - accuracy: 0.9667 - val_loss: 0.2008 - val_accuracy: 0.9458
Epoch 139/150
21/21 [=====] - 16s 772ms/step - loss:
0.1091 - accuracy: 0.9545 - val_loss: 0.2511 - val_accuracy: 0.9458
Epoch 140/150
21/21 [=====] - 16s 759ms/step - loss:
0.0956 - accuracy: 0.9606 - val_loss: 0.2112 - val_accuracy: 0.9157
Epoch 141/150
21/21 [=====] - 16s 761ms/step - loss:
0.1170 - accuracy: 0.9470 - val_loss: 0.2441 - val_accuracy: 0.9398
Epoch 142/150
21/21 [=====] - 16s 747ms/step - loss:
0.0923 - accuracy: 0.9652 - val_loss: 0.3449 - val_accuracy: 0.9337

Epoch 143/150

21/21 [=====] - 16s 753ms/step - loss:
0.1012 - accuracy: 0.9652 - val_loss: 0.5842 - val_accuracy: 0.8976

Epoch 144/150

21/21 [=====] - ETA: 0s - loss: 0.1155 -
accuracy: 0.9576WARNING:absl:Found untraced functions such as
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op, _jit_compiled_convolution_op,
_jit_compiled_convolution_op while saving (showing 5 of 6). These
functions will not be directly callable after loading.

21/21 [=====] - 17s 820ms/step - loss:
0.1155 - accuracy: 0.9576 - val_loss: 0.2764 - val_accuracy: 0.9578

Epoch 145/150

21/21 [=====] - 16s 767ms/step - loss:
0.0840 - accuracy: 0.9712 - val_loss: 0.2535 - val_accuracy: 0.9217

Epoch 146/150

21/21 [=====] - 16s 756ms/step - loss:
0.0756 - accuracy: 0.9758 - val_loss: 0.3914 - val_accuracy: 0.9096

Epoch 147/150

21/21 [=====] - 16s 754ms/step - loss:
0.0944 - accuracy: 0.9682 - val_loss: 0.2952 - val_accuracy: 0.9458

Epoch 148/150

21/21 [=====] - 16s 757ms/step - loss:
0.1004 - accuracy: 0.9652 - val_loss: 0.3896 - val_accuracy: 0.9398

Epoch 149/150

21/21 [=====] - 17s 837ms/step - loss:
0.0816 - accuracy: 0.9667 - val_loss: 0.5512 - val_accuracy: 0.9398

Epoch 150/150

21/21 [=====] - 16s 767ms/step - loss:
0.0723 - accuracy: 0.9758 - val_loss: 0.4174 - val_accuracy: 0.9398

10.10. Load the saved model

```
saved_model = load_model(saved_models_path +  
"/saved_model")
```

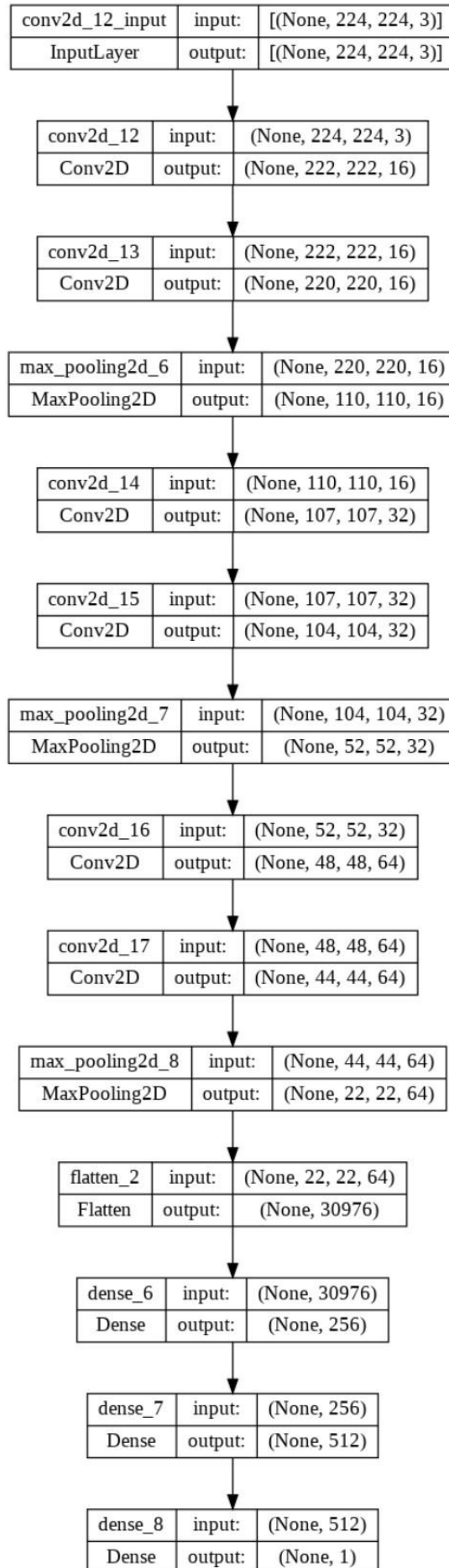
```
saved_model.summary()
```

```
Model: "sequential_2"
```

Layer (type)	Output Shape	Param #
conv2d_12 (Conv2D)	(None, 222, 222, 16)	448
conv2d_13 (Conv2D)	(None, 220, 220, 16)	2320
max_pooling2d_6 (MaxPooling2D)	(None, 110, 110, 16)	0
conv2d_14 (Conv2D)	(None, 107, 107, 32)	8224
conv2d_15 (Conv2D)	(None, 104, 104, 32)	16416
max_pooling2d_7 (MaxPooling2D)	(None, 52, 52, 32)	0
conv2d_16 (Conv2D)	(None, 48, 48, 64)	51264
conv2d_17 (Conv2D)	(None, 44, 44, 64)	102464
max_pooling2d_8 (MaxPooling2D)	(None, 22, 22, 64)	0
flatten_2 (Flatten)	(None, 30976)	0
dense_6 (Dense)	(None, 256)	7930112
dense_7 (Dense)	(None, 512)	131584
dense_8 (Dense)	(None, 1)	513

Total params: 8,243,345
Trainable params: 8,243,345
Non-trainable params: 0

```
from tensorflow.keras.utils import plot_model  
plot_model(saved_model, show_shapes= True )
```



10.11. Evaluate on Training Data, Augmented Training Data and Test Data

```
saved_model.evaluate(train_data)
```

```
21/21 [=====] - 8s 373ms/step - loss: 0.0522 - accuracy: 0.9833  
[0.052219975739717484, 0.9833333492279053]
```

```
saved_model.evaluate(train_data_augmented)
```

```
21/21 [=====] - 13s 637ms/step - loss: 0.0611 - accuracy: 0.9833  
[0.06107011437416077, 0.9833333492279053]
```

```
saved_model.evaluate(test_data)
```

```
6/6 [=====] - 2s 337ms/step - loss: 0.2764 - accuracy: 0.9578  
[0.2763960361480713, 0.9578313231468201]
```

10.12. Evaluate on custom images over the web

```
custom_birds = "/content/drive/MyDrive/custom_test/birds"  
custom_drones =  
"/content/drive/MyDrive/custom_test/drones" birds_images =  
os.listdir(custom_birds) drones_images =  
os.listdir(custom_drones) walk_through_dir(custom_birds)
```

```
There are 0 directories and 4 images in  
'/content/drive/MyDrive/custom_test/birds'.
```

```
pred_and_plot_directory(saved_model, custom_birds)
```


Prediction: birds
Probability: 0.6749



Prediction: birds
Probability: 1.0000



Prediction: birds
Probability: 0.9592



Prediction: birds
Probability: 0.8488



```
walk_through_dir(custom_drones)
```

There are 0 directories and 4 images in
'/content/drive/MyDrive/custom_test/drones'.

```
pred_and_plot_directory(saved_model, custom_drones)
```

Prediction: drones
Probability: 1.0000



Prediction: drones
Probability: 0.9982



Prediction: drones
Probability: 0.9582



Prediction: drones
Probability: 0.6235



11.TEST CASES AND TEST RESULTS

Test Id	Test Case Title	Test Condition	System Behaviour	Expected Result
T01	TestCase 1	Bird in clear sky	Birds	Birds
T02	TestCase 2	Drone in clear sky	Drones	Drones
T03	TestCase 3	Birds among the clouds	Birds	Birds

12. CONCLUSION

We have successfully developed an architecture for a binary image classification model that classifies whether an unidentified flying object is a bird or a drone, trained it on roughly 400 images of birds and drones each and tested the performance of the model on previously unseen images (test images) and also evaluated on custom images downloaded from the web. As does every effort being made in the universe, this project also has a lot of scope for enhancement. This is a simple and basic project that we've built and there are a lot of enhancements and improvements that can be made due to the great significance and importance of the application and functionality that this project aims at providing.

The following are a list of enhancements that can be made to further improve the functionality of this project:

1. **Accuracy:** Right now, the accuracy of the Artificial Neural Network (ANN) model that is trained is roughly 91% which is not a really an excellent value. We can perform further training to the model by providing more images for the model to learn on.
2. **Live Camera View:** Instead of capturing the image and providing it as the input to our trained model, we can use live video feed from a high quality camera to focus on the flying object and use this live feed with concepts of Computer Vision along with trained Artificial Neural Network (ANN) model that is trained.
3. **Fire an identified drone:** The model can be used as the brain of a laser gun that can focus properly on the d—rone and take down an identified drone eliminating any potential threat.

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